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# User Manual

for MPC560XB LIN Driver

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# Chapter 1

## Revision History

**Table 1-1. Revision History**

| Revision | Date        | Author                 | Description                                     |
|----------|-------------|------------------------|-------------------------------------------------|
| 1.0      | 31-Jan-2012 | Giuseppe Stefano Fazio | First version for Beta 0.9.0 release            |
| 1.1      | 06-May-2013 | Giuseppe Stefano Fazio | Updated the manual for Bolero RTM 1.0.0 release |
| 1.2      | 06-May-2014 | Giuseppe Stefano Fazio | Updated the manual for Bolero RTM 1.0.1 release |



# Chapter 2

## Introduction

This User Manual describes Freescale Semiconductor AUTOSAR Local Interconnect Network ( Lin ) for MPC560XB .

AUTOSAR Lin driver configuration parameters and deviations from the specification are described in Lin Driver chapter of this document. AUTOSAR Lin driver requirements and APIs are described in the AUTOSAR Lin driver software specification document.

### 2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of Freescale Semiconductor .

**Table 2-1. MPC560XB Derivatives**

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Freescale Semiconductor | mpc5601dxlh_lqfp64<br>mpc5601dxll_lqfp100<br>mpc5602dxlh_lqfp64<br>mpc5602dxll_lqfp100<br>mpc5602bxlh_lqfp64<br>mpc5602bxll_lqfp100<br>mpc5602bxlq_lqfp144<br>mpc5602cxlh_lqfp64<br>mpc5602cxll_lqfp100<br>mpc5603bxlh_lqfp64<br>mpc5603bxll_lqfp100<br>mpc5603bxlq_lqfp144<br>mpc5603cxlh_lqfp64<br>mpc5603cxll_lqfp100<br>mpc5604bxlh_lqfp64<br>mpc5604bxll_lqfp100<br>mpc5604bxlq_lqfp144<br>mpc5604bxmg_mapbga208<br>mpc5604cxlh_lqfp64<br>mpc5604cxll_lqfp100<br>mpc5605bxll_lqfp100<br>mpc5605bxlq_lqfp144<br>mpc5605bxlu_lqfp176<br>mpc5606bkll_lqfp100<br>mpc5606bxlq_lqfp144 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Table 2-1. MPC560XB Derivatives**

|  |                                                                     |
|--|---------------------------------------------------------------------|
|  | mpc5606bxlu_lqfp176<br>mpc5607bxlu_lqfp176<br>mpc5607bxmg_mapbga208 |
|--|---------------------------------------------------------------------|

All of the above microcontroller devices are collectively named as MPC560XB .

## 2.2 Overview

**AUTOSAR (AUTomotive Open System ARchitecture)** is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

### AUTOSAR

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

## 2.3 About this Manual

This Technical Reference employs the following typographical conventions:

**Boldface** type: Bold is used for important terms, notes and warnings.

*Italic* font: Italic typeface is used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

### Note

This is a note.

## 2.4 Acronyms and Definitions

**Table 2-2. Acronyms and Definitions**

| Term    | Definition                          |
|---------|-------------------------------------|
| API     | Application Programming Interface   |
| ASM     | Assembler                           |
| AUTOSAR | Automotive Open System Architecture |
| BSMI    | Basic Software Make file Interface  |
| C/CPP   | C and C++ Source Code               |
| DEM     | Diagnostic Event Manager            |
| DET     | Development Error Tracer            |
| EcuM    | ECU state Manager                   |
| GUI     | Graphical User Interface            |
| ISR     | Interrupt Service Routine           |
| LIN     | Local Interconnect Network          |
| MCU     | Micro Controller Unit               |
| N/A     | Not Applicable                      |
| OS      | Operating System                    |
| VLE     | Variable Length Encoding            |

## 2.5 Reference List

**Table 2-3. Reference List**

| # | Title                                                          | Version                  |
|---|----------------------------------------------------------------|--------------------------|
| 1 | AUTOSAR 4.0 Rev0003Lin Driver Software Specification Document. | V1.5.0                   |
| 2 | MPC5607B Reference Manual                                      | Rev. 7.2, May 2012       |
| 3 | Bolero 1.5M cut 2.2 Errata Sheet<br>(MPC5607B_1M03Y)           | Rev. 12Dec2013           |
| 4 | MPC5604B Reference Manual                                      | Rev. 8.2, September 2013 |
| 5 | Bolero 512k cut 2.3 Errata Sheet<br>(MPC5604B_2M27V)           | Rev. 26 FEB 2014         |
| 6 | 5606B Reference Manual                                         | Rev. 2, May 2014         |
| 7 | 5602D Reference Manual                                         | Rev. 4.2, Dec 2013       |
| 8 | Bolero 256k cut 1.1 Errata Sheet<br>(MPC5602D_1M18Y)           | Rev. , 12 DEC 2013       |



## Chapter 3 Driver

### 3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.0 Rev0003Lin Driver Software Specification document (See Table [Reference List](#) ).

### 3.2 Driver Design Summary

The LIN driver is part of the microcontroller abstraction layer (MCAL), performs the hardware access and offers a hardware independent API to the upper layer.

The only upper layer, which has access to the LIN driver, is the LIN Interface.

A LIN driver can support more than one channel.

This means that the LIN driver can handle one or more LIN channels as long as they belong to the same LIN hardware unit.

The LIN Driver for MPC560XB, uses the LINFlex on-chip hardware module which provides special support for the LIN protocol.

It can be used to automate most tasks of a LIN master.

It is possible to transmit entire frames (or sequences of frames) and receive data from LIN slaves without any CPU intervention.

The LIN physical interface should be connected to the LINFlex module pins in order to get the LIN bus voltage levels.

The MPC560XB contains up to ten blocks.

The LINFlex has the following major features:

- Supports LIN Master mode, LIN Slave mode and UART mode
- LIN state machine compliant to LIN1.3, 2.0, and 2.1 specifications
- Manages LIN frame transmission and reception without CPU intervention
- Autonomous LIN frame handling
- Message buffer to store as many as 8 data bytes
- Supports messages as long as 64 bytes
- Detection and flagging of LIN errors (Sync field, delimiter, ID parity, bit framing, checksum and Time-out errors)
- Classic or extended checksum calculation
- Configurable break duration of up to 36-bit times
- Programmable baud rate prescalers (13-bit mantissa, 4-bit fractional)
- Diagnostic features (Loop back, LIN bus stuck dominant detection)
- Interrupt driven operation with 16 interrupt sources

### 3.3 Deviation from Requirements

The driver deviates from the AUTOSAR Lin Driver software specification in some places. Table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the Lin driver. Table [Table 3-1](#) provides Status column description.

**Table 3-1. Deviations Status Column Description**

| Term | Definition                  |
|------|-----------------------------|
| N/A  | Not available               |
| N/T  | Not testable                |
| N/S  | Out of scope                |
| N/R  | Unclear Requirement         |
| N/I  | Not implemented             |
| N/F  | Not fully implemented       |
| I/D  | Implemented with Deviations |

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the driver.

**Table 3-2. Driver Deviations Table**

| Requirement | Status | Description                                                                                                                                                   | Notes                                                                                                                                                                                       |
|-------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LIN109      | N/R    | If development error detection for the LIN module is enabled: the function Lin_CheckWakeup shall raise the development error LIN_E_STATE_TRANSITION if no LIN | The LIN driver has to check in any case, all the LIN channels to found if no LIN channel of the driver is in the LIN_CH_SLEEP state: it means that the time benefits gained by an immediate |



**Table 3-2. Driver Deviations Table**

| Requirement | Status | Description                                         | Notes                                                                              |
|-------------|--------|-----------------------------------------------------|------------------------------------------------------------------------------------|
|             |        | channel of the driver is in the LIN_CH_SLEEP state. | check on the addressed Channel will be lost by a second check on all the channels. |

## 3.4 Runtime Errors

The driver generates the following DEM errors at runtime.

**Table 3-3. Runtime Errors**

| Function                | Error Code    | Condition triggering the error                                              |
|-------------------------|---------------|-----------------------------------------------------------------------------|
| Lin_GoToSleep()         | LIN_E_TIMEOUT | Timeout caused by hardware error waiting for cancellation of current frame. |
| Lin_GoToSleepInternal() | LIN_E_TIMEOUT | Timeout caused by hardware error waiting for cancellation of current frame. |
| Lin_SendFrame()         | LIN_E_TIMEOUT | Timeout caused by hardware error waiting for cancellation of current frame. |

## 3.5 Software specification

The following sections contains driver software specifications.

### 3.5.1 Define Reference

Constants supported by the Lin driver.

#### 3.5.1.1 Define LIN\_BIT\_ERROR

Interrupt Errors conditions.

##### Details:

Bit error on a channel:

- During response field transmission (Slave and Master modes);
- During header transmission (in Master mode).

**Implements:** DLIN02003

**Table 3-4. Define LIN\_BIT\_ERROR Description**

|                    |               |
|--------------------|---------------|
| <b>Name</b>        | LIN_BIT_ERROR |
| <b>Initializer</b> | 0x01U         |

### 3.5.1.2 Define LIN\_BREAK\_DELIMITER\_ERROR

Interrupt Errors conditions.

**Details:**

Break Delimiter too short (< 1 bit).

**Implements:** DLIN02003

**Table 3-5. Define LIN\_BREAK\_DELIMITER\_ERROR Description**

|                    |                           |
|--------------------|---------------------------|
| <b>Name</b>        | LIN_BREAK_DELIMITER_ERROR |
| <b>Initializer</b> | 0x04U                     |

### 3.5.1.3 Define LIN\_BUFFER\_OVER\_RUN\_ERROR

Interrupt Errors conditions.

**Details:**

New data byte is received on a channel and the buffer full flag is not cleared.

**Implements:** DLIN02003

**Table 3-6. Define LIN\_BUFFER\_OVER\_RUN\_ERROR Description**

|                    |                           |
|--------------------|---------------------------|
| <b>Name</b>        | LIN_BUFFER_OVER_RUN_ERROR |
| <b>Initializer</b> | 0x07U                     |

### 3.5.1.4 Define LIN\_CH\_NOT\_READY\_STATE

LIN Channel states.

#### Details:

The individual channel is not ready to process a frame.

**Implements:** DLIN02007

**Table 3-7. Define LIN\_CH\_NOT\_READY\_STATE Description**

|                    |                        |
|--------------------|------------------------|
| <b>Name</b>        | LIN_CH_NOT_READY_STATE |
| <b>Initializer</b> | 0x04U                  |

### 3.5.1.5 Define LIN\_CH\_OPERATIONAL\_STATE

LIN Channel states.

#### Details:

The individual channel has been initialized (using at least one statically configured data set) and is able to participate in the LIN cluster.

**Implements:** DLIN02004, DLIN09002

**Table 3-8. Define LIN\_CH\_OPERATIONAL\_STATE Description**

|                    |                          |
|--------------------|--------------------------|
| <b>Name</b>        | LIN_CH_OPERATIONAL_STATE |
| <b>Initializer</b> | 0x03U                    |

### 3.5.1.6 Define LIN\_CH\_READY\_STATE

LIN Channel states.

#### Details:

The individual channel is ready to process a frame.

**Implements:** DLIN02007

**Table 3-9. Define LIN\_CH\_READY\_STATE Description**

|                    |                    |
|--------------------|--------------------|
| <b>Name</b>        | LIN_CH_READY_STATE |
| <b>Initializer</b> | 0x05U              |

### 3.5.1.7 Define LIN\_CH\_RECEIVE\_NOTHING\_STATE

LIN Channel states.

**Details:**

State after the LIN frame header was correctly sent.

**Implements:** DLIN02007

**Table 3-10. Define LIN\_CH\_RECEIVE\_NOTHING\_STATE Description**

|                    |                              |
|--------------------|------------------------------|
| <b>Name</b>        | LIN_CH_RECEIVE_NOTHING_STATE |
| <b>Initializer</b> | 0x08U                        |

### 3.5.1.8 Define LIN\_CH\_SLEEP\_STATE

LIN Channel states.

**Details:**

The detection of a wake-up pulse is enabled. The LIN hardware is into a low power mode if such a mode is provided by the hardware.

**Implements:** DLIN02004

**Table 3-11. Define LIN\_CH\_SLEEP\_STATE Description**

|                    |                    |
|--------------------|--------------------|
| <b>Name</b>        | LIN_CH_SLEEP_STATE |
| <b>Initializer</b> | 0x02U              |

### 3.5.1.9 Define LIN\_CHECKSUM\_ERROR

Interrupt Errors conditions.

#### Details:

Checksum error on a channel.

**Implements:** DLIN02003

**Table 3-12. Define LIN\_CHECKSUM\_ERROR Description**

|                    |                    |
|--------------------|--------------------|
| <b>Name</b>        | LIN_CHECKSUM_ERROR |
| <b>Initializer</b> | 0x02U              |

### 3.5.1.10 Define LIN\_E\_INVALID\_CHANNEL

API service used with an invalid or inactive channel parameter.

#### Details:

The LIN Driver module shall report the development error "LIN\_E\_INVALID\_CHANNEL (0x02)", when API Service used with an invalid or inactive channel parameter.

**Implements:** DLIN02002

**Table 3-13. Define LIN\_E\_INVALID\_CHANNEL Description**

|                    |                       |
|--------------------|-----------------------|
| <b>Name</b>        | LIN_E_INVALID_CHANNEL |
| <b>Initializer</b> | ((uint8)0x02U)        |

### 3.5.1.11 Define LIN\_E\_INVALID\_POINTER

API service called with invalid configuration pointer.

#### Details:

The LIN Driver module shall report the development error "LIN\_E\_INVALID\_POINTER (0x03)", when API Service is called with invalid configuration pointer.

**Implements:** DLIN02002

**Table 3-14. Define LIN\_E\_INVALID\_POINTER Description**

|                    |                       |
|--------------------|-----------------------|
| <b>Name</b>        | LIN_E_INVALID_POINTER |
| <b>Initializer</b> | ((uint8)0x03U)        |

### 3.5.1.12 Define LIN\_E\_PARAM\_POINTER

API service called with a NULL pointer.

**Details:**

The LIN Driver module shall report the development error "LIN\_E\_PARAM\_POINTER (0x05)", when API Service is called with a NULL pointer. In case of this error, the API service shall return immediately without any further action, beside reporting this development error.

**Implements:** DLIN02002

**Table 3-15. Define LIN\_E\_PARAM\_POINTER Description**

|                    |                     |
|--------------------|---------------------|
| <b>Name</b>        | LIN_E_PARAM_POINTER |
| <b>Initializer</b> | ((uint8)0x05U)      |

### 3.5.1.13 Define LIN\_E\_STATE\_TRANSITION

Invalid state transition for the current state.

**Details:**

The LIN Driver module shall report the development error "LIN\_E\_STATE\_TRANSITION (0x04)", when Invalid state transition occurs from the current state.

**Implements:** DLIN02002

**Table 3-16. Define LIN\_E\_STATE\_TRANSITION Description**

|                    |                        |
|--------------------|------------------------|
| <b>Name</b>        | LIN_E_STATE_TRANSITION |
| <b>Initializer</b> | ((uint8)0x04U)         |

### 3.5.1.14 Define LIN\_E\_UNINIT

API service used without module initialization.

#### Details:

The LIN Driver module shall report the development error "LIN\_E\_UNINIT (0x00)", when the API Service is used without module initialization.

**Implements:** DLIN02002

**Table 3-17. Define LIN\_E\_UNINIT Description**

|                    |                |
|--------------------|----------------|
| <b>Name</b>        | LIN_E_UNINIT   |
| <b>Initializer</b> | ((uint8)0x00U) |

### 3.5.1.15 Define LIN\_FRAMING\_ERROR

Interrupt Errors conditions.

#### Details:

Invalid stop bit:

- During reception of any data in the response field (Slave and Master modes);
- During reception of Synch or Identifier Field (Slave mode).

**Implements:** DLIN02003

**Table 3-18. Define LIN\_FRAMING\_ERROR Description**

|                    |                   |
|--------------------|-------------------|
| <b>Name</b>        | LIN_FRAMING_ERROR |
| <b>Initializer</b> | 0x06U             |

### 3.5.1.16 Define LIN\_IDENTIFIER\_PARITY\_ERROR

Interrupt Errors conditions.

#### Details:

Parity error.

**Implements:** DLIN02003

**Table 3-19. Define LIN\_IDENTIFIER\_PARITY\_ERROR  
Description**

|                    |                             |
|--------------------|-----------------------------|
| <b>Name</b>        | LIN_IDENTIFIER_PARITY_ERROR |
| <b>Initializer</b> | 0x05U                       |

### 3.5.1.17 Define LIN\_INIT

LIN driver states.

#### Details:

The LIN\_INIT state indicates that the LIN driver has been initialized, making each available channel ready for service.

**Implements:** DLIN02006, DLIN09001

**Table 3-20. Define LIN\_INIT Description**

|                    |          |
|--------------------|----------|
| <b>Name</b>        | LIN_INIT |
| <b>Initializer</b> | 0x02U    |

### 3.5.1.18 Define LIN\_NO\_ERROR

Interrupt Errors conditions.

#### Details:



No error occurred on a channel.

**Implements:** DLIN02003

**Table 3-21. Define LIN\_NO\_ERROR Description**

|                    |              |
|--------------------|--------------|
| <b>Name</b>        | LIN_NO_ERROR |
| <b>Initializer</b> | 0x00U        |

### 3.5.1.19 Define LIN\_NOISE\_ERROR

Interrupt Errors conditions.

**Details:**

Noise detected on a received character.

**Implements:** DLIN02003

**Table 3-22. Define LIN\_NOISE\_ERROR Description**

|                    |                 |
|--------------------|-----------------|
| <b>Name</b>        | LIN_NOISE_ERROR |
| <b>Initializer</b> | 0x08U           |

### 3.5.1.20 Define LIN\_RX\_COMPLETE\_STATE

LIN Channel states.

**Details:**

LIN frame was received; no errors.

**Implements:** DLIN02007

**Table 3-23. Define LIN\_RX\_COMPLETE\_STATE Description**

|                    |                       |
|--------------------|-----------------------|
| <b>Name</b>        | LIN_RX_COMPLETE_STATE |
| <b>Initializer</b> | 0x07U                 |

### 3.5.1.21 Define LIN\_SYNC\_FIELD\_ERROR

Interrupt Errors conditions.

#### Details:

Inconsistent Synch Field.

**Implements:** DLIN02003

**Table 3-24. Define LIN\_SYNC\_FIELD\_ERROR Description**

|                    |                      |
|--------------------|----------------------|
| <b>Name</b>        | LIN_SYNC_FIELD_ERROR |
| <b>Initializer</b> | 0x03U                |

### 3.5.1.22 Define LIN\_TIMEOUT\_ERROR

Interrupt Errors conditions.

#### Details:

Header or Response timeout detected.

**Implements:** DLIN02003

**Table 3-25. Define LIN\_TIMEOUT\_ERROR Description**

|                    |                   |
|--------------------|-------------------|
| <b>Name</b>        | LIN_TIMEOUT_ERROR |
| <b>Initializer</b> | 0x09U             |

### 3.5.1.23 Define LIN\_TX\_COMPLETE\_STATE

LIN Channel states.

#### Details:

LIN frame was sent; no errors.

**Implements:** DLIN02007

**Table 3-26. Define LIN\_TX\_COMPLETE\_STATE Description**

|                    |                       |
|--------------------|-----------------------|
| <b>Name</b>        | LIN_TX_COMPLETE_STATE |
| <b>Initializer</b> | 0x06U                 |

### 3.5.1.24 Define LIN\_TX\_MASTER\_RES\_COMMAND

Commands IDs.

#### Details:

Tx frame is a master frame (response is provided by master).

**Implements:** DLIN02005

**Table 3-27. Define LIN\_TX\_MASTER\_RES\_COMMAND Description**

|                    |                           |
|--------------------|---------------------------|
| <b>Name</b>        | LIN_TX_MASTER_RES_COMMAND |
| <b>Initializer</b> | 0x01U                     |

### 3.5.1.25 Define LIN\_TX\_NO\_COMMAND

Commands IDs.

#### Details:

No tx master command pending.

**Implements:** DLIN02005

**Table 3-28. Define LIN\_TX\_NO\_COMMAND Description**

|                    |                   |
|--------------------|-------------------|
| <b>Name</b>        | LIN_TX_NO_COMMAND |
| <b>Initializer</b> | 0x04U             |

### 3.5.1.26 Define LIN\_TX\_SLAVE\_RES\_COMMAND

Commands IDs.

**Details:**

Tx frame is a slave frame (response is provided by slave).

**Implements:** DLIN02005

**Table 3-29. Define LIN\_TX\_SLAVE\_RES\_COMMAND Description**

|                    |                          |
|--------------------|--------------------------|
| <b>Name</b>        | LIN_TX_SLAVE_RES_COMMAND |
| <b>Initializer</b> | 0x02U                    |

### 3.5.1.27 Define LIN\_TX\_SLEEP\_COMMAND

Commands IDs.

**Details:**

Tx frame is a sleep command frame.

**Implements:** DLIN02005

**Table 3-30. Define LIN\_TX\_SLEEP\_COMMAND Description**

|                    |                      |
|--------------------|----------------------|
| <b>Name</b>        | LIN_TX_SLEEP_COMMAND |
| <b>Initializer</b> | 0x03U                |

### 3.5.1.28 Define LIN\_UNINIT

LIN driver states.

**Details:**

The state LIN\_UNINIT means that the Lin module has not been initialized yet and cannot be used.

**Implements:** DLIN02006, DLIN09003

**Table 3-31. Define LIN\_UNINIT Description**

|                    |            |
|--------------------|------------|
| <b>Name</b>        | LIN_UNINIT |
| <b>Initializer</b> | 0x01U      |

### 3.5.1.29 Define LIN\_DEV\_ERROR\_DETECT

Switches the Development Error Detection and Notification ON or OFF.

**Implements:** DLIN03001

**Table 3-32. Define LIN\_DEV\_ERROR\_DETECT Description**

|                    |                      |
|--------------------|----------------------|
| <b>Name</b>        | LIN_DEV_ERROR_DETECT |
| <b>Initializer</b> | (STD_ON)             |

### 3.5.1.30 Define LIN\_DISABLE\_DEM\_REPORT\_ERROR\_STATUS

Switches the Production Error Detection and Notification OFF.

**Implements:** DLIN03016

**Table 3-33. Define LIN\_DISABLE\_DEM\_REPORT\_ERROR\_STATUS Description**

|                    |                                     |
|--------------------|-------------------------------------|
| <b>Name</b>        | LIN_DISABLE_DEM_REPORT_ERROR_STATUS |
| <b>Initializer</b> | (STD_ON)                            |

### 3.5.1.31 Define LIN\_DUAL\_CLOCK\_MODE

Enable Non-Autosar API for Dual-Clock support.

**Details:**

Enable/disable API Lin\_SetClockMode() to set the clock to be used by the LIN driver (Normal clock: default mode; Alternate clock: when the driver is in Low-Power mode). This can be set to STD\_ON only if it is activated from xdm file (Lin.NonAutosar.LinEnableDualClockMode = STD\_ON).

**Implements:** DLIN03002

**Table 3-34. Define LIN\_DUAL\_CLOCK\_MODE Description**

|                    |                     |
|--------------------|---------------------|
| <b>Name</b>        | LIN_DUAL_CLOCK_MODE |
| <b>Initializer</b> | (STD_OFF)           |

### 3.5.1.32 Define LIN\_GLOBAL\_WAKEUP\_SUPPORT

At least 1 LIN channel has WAKEUP support enabled.

**Details:**

At least 1 LIN channel has WAKEUP support enabled.

**Implements:** DLIN03003

**Table 3-35. Define LIN\_GLOBAL\_WAKEUP\_SUPPORT Description**

|                    |                           |
|--------------------|---------------------------|
| <b>Name</b>        | LIN_GLOBAL_WAKEUP_SUPPORT |
| <b>Initializer</b> |                           |

### 3.5.1.33 Define LIN\_HW\_MAX\_AVAILABLE\_MODULES

Total number of available hardware lin channels.

**Implements:** DLIN03004

**Table 3-36. Define LIN\_HW\_MAX\_AVAILABLE\_MODULES Description**

|                    |                              |
|--------------------|------------------------------|
| <b>Name</b>        | LIN_HW_MAX_AVAILABLE_MODULES |
| <b>Initializer</b> | 10U                          |

### 3.5.1.34 Define LIN\_HW\_MAX\_MODULES

No of Channels configured.

**Implements:** DLIN03005

**Table 3-37. Define LIN\_HW\_MAX\_MODULES Description**

|                    |                    |
|--------------------|--------------------|
| <b>Name</b>        | LIN_HW_MAX_MODULES |
| <b>Initializer</b> | 2U                 |

### 3.5.1.35 Define LIN\_MAX\_DATA\_LENGTH

Max data length of the LIN SDU buffer to be returned.

**Implements:** DLIN03006

**Table 3-38. Define LIN\_MAX\_DATA\_LENGTH Description**

|                    |                     |
|--------------------|---------------------|
| <b>Name</b>        | LIN_MAX_DATA_LENGTH |
| <b>Initializer</b> | 8U                  |

### 3.5.1.36 Define LIN\_TIMEOUT\_LOOPS

Number of loops before returning LIN\_E\_TIMEOUT.

**Implements:** DLIN03008

**Table 3-39. Define LIN\_TIMEOUT\_LOOPS Description**

|                    |                   |
|--------------------|-------------------|
| <b>Name</b>        | LIN_TIMEOUT_LOOPS |
| <b>Initializer</b> | 1000U             |

### 3.5.1.37 Define LIN\_VERSION\_INFO\_API

Support for version info API.

#### Details:

Switches the `Lin_GetVersionInfo()` API ON or OFF.

**Implements:** DLIN03009

**Table 3-40. Define LIN\_VERSION\_INFO\_API Description**

|                    |                      |
|--------------------|----------------------|
| <b>Name</b>        | LIN_VERSION_INFO_API |
| <b>Initializer</b> | (STD_ON)             |

### 3.5.1.38 Define LinConf\_LinChannelConfiguration\_LinChannel\_0

Symbolic names for configured channels (as per `ecuc_sws_2108`).

#### Details:

Symbolic names for configured channels (as per `ecuc_sws_2108`).

**Implements:** DLIN03011

**Table 3-41. Define LinConf\_LinChannelConfiguration\_LinChannel\_0 Description**

|                    |                                              |
|--------------------|----------------------------------------------|
| <b>Name</b>        | LinConf_LinChannelConfiguration_LinChannel_0 |
| <b>Initializer</b> | 0                                            |

## 3.5.2 Macros Reference

Macros supported by the Lin driver.

### 3.5.2.1 Macro LIN\_GET\_BASE\_ADDRESS

LINFLEX Registers macro definition.

#### Details:



LINFLEX macro to access base addresses of LINFlex IPs.

**Violates:** include statements in a file should only be preceded by other preprocessor directives or comments.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LIN_GET_BASE_ADDRESS(offset)`

### 3.5.2.2 Macro LINFLEX\_BDRL

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register Buffer Data Register Least Significant (BDRL).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_BDRL(offset)`

### 3.5.2.3 Macro LINFLEX\_BDRM

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register Buffer Data Register Most Significant (BDRM).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_BDRM(offset)`

### 3.5.2.4 Macro LINFLEX\_BIDR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register Buffer Identifier Register (BIDR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_BIDR(offset)

### 3.5.2.5 Macro LINFLEX\_EXIT\_NORMAL\_MODE\_TO\_INIT\_MODE

LINCR1 registers - Exit Normal Mode Enters Init Mode.

**Details:**

Exit the Normal mode and enters the Sleep mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_EXIT\_NORMAL\_MODE\_TO\_INIT\_MODE(offset)

### 3.5.2.6 Macro LINFLEX\_LINCFR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Checksum Field Register (LINCFR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LINCFR(offset)

### 3.5.2.7 Macro LINFLEX\_LINCR1

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Control Register 1 (LINCR1).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LINCR1(offset)

### 3.5.2.8 Macro LINFLEX\_LINCR2

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Control Register 2 (LINCR2).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LINCR2(offset)

### 3.5.2.9 Macro LINFLEX\_LINESR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Error Status Register (LINESR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LINESR(offset)

### 3.5.2.10 Macro LINFLEX\_LINFBRR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Fractional Baud Rate Register (LINFBRR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LINFBR(offset)`

### 3.5.2.11 Macro LINFLEX\_LINIBRR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Integer Baud Rate Register (LINIBRR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LINIBRR(offset)`

### 3.5.2.12 Macro LINFLEX\_LINIER

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Interrupt Enable Register (LINIER).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LINIER(offset)`

### 3.5.2.13 Macro LINFLEX\_LINOCR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Output Compare Register (LINOCR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LINOCR(offset)`

### 3.5.2.14 Macro LINFLEX\_LINSR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Status Register (LINSR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LINSR(offset)

### 3.5.2.15 Macro LINFLEX\_LINTCSR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Time-Out Control Status Register (LINTCSR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LINTCSR(offset)

### 3.5.2.16 Macro LINFLEX\_LINTOCR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register LIN Time-Out Control Register (LINTOCR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LINTOCR(offset)

### 3.5.2.17 Macro LINFLEX\_LLM\_7\_BITS\_DATA\_PARITY\_Set

UARTCR registers - Word Length in UART mode.

**Details:**

Set the 8-bit data (or 9-bit if PCE is set) in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_7_BITS_DATA_PARITY_Set(offset)`

### **3.5.2.18 Macro LINFLEX\_LLM\_8\_BITS\_DATA\_PARITY\_Set**

UARTCR registers - Word Length in UART mode.

**Details:**

Set the 7-bit data + parity bit in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_8_BITS_DATA_PARITY_Set(offset)`

### **3.5.2.19 Macro LINFLEX\_LLM\_ABRQ\_Read**

LINCR2 registers - Read Abort Request bits.

**Details:**

Read the Abort Request bits.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_ABRQ_Read(offset)`

### **3.5.2.20 Macro LINFLEX\_LLM\_ABRQ\_Set**

LINCR2 registers - Abort Request.

**Details:**

Abort the current transmission.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_ABRQ_Set(offset)`

### 3.5.2.21 Macro LINFLEX\_LLM\_AWUM\_Disable

LINCR1 registers - Automatic Wake-Up Mode Disable.

**Details:**

The Sleep mode is exited on software request by clearing the SLEEP bit of the LINCR register.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_AWUM_Disable(offset)`

### 3.5.2.22 Macro LINFLEX\_LLM\_AWUM\_Enable

LINCR1 registers - Automatic Wake-Up Mode Enable.

**Details:**

The Sleep mode is exited automatically by hardware on RX dominant state detection. The SLEEP bit of the LINCR register is cleared by hardware whenever WUF bit in LINSR is set.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_AWUM_Enable(offset)`

### 3.5.2.23 Macro LINFLEX\_LLM\_BDEF\_Clear

LINESR registers - Break Delimiter Error Flag.

**Details:**

This macro clears the Break Delimiter Error Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BDEF_Clear(offset)`

### 3.5.2.24 Macro LINFLEX\_LLM\_BDEF\_Read

LINESR registers - Break Delimiter Error Flag.

**Details:**

This macro read the bit indicating to the software that received Break Delimiter is too short (less than one bit time).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BDEF_Read(offset)`

### 3.5.2.25 Macro LINFLEX\_LLM\_BDRL\_Read

BDRL registers - Read all the BDRL register.

**Details:**

Read all the BDRL register (Data Bytes number 3,2,1 and 0 of the data field).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BDRL_Read(offset)`

### 3.5.2.26 Macro LINFLEX\_LLM\_BDRL\_Write

BDRL registers - Write all the BDRL register.

**Details:**

Write all the BDRL register (Data Bytes number 3,2,1 and 0 of the data field).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.



**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_BDRL_Write(offset,value)`

### 3.5.2.27 Macro LINFLEX\_LLM\_BDRM\_Read

BDRM registers - Read all the BDRM register.

#### **Details:**

Read all the BDRM register (Data Bytes number 7,6,5 and 4 of the data field).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BDRM_Read(offset)`

### 3.5.2.28 Macro LINFLEX\_LLM\_BDRM\_Write

BDRM registers - Write all the BDRM register.

#### **Details:**

Write all the BDRM register (Data Bytes number 7,6,5 and 4 of the data field).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_BDRM_Write(offset,value)`

### 3.5.2.29 Macro LINFLEX\_LLM\_BEFClear

LINEAR registers - Bit Error Flag.

#### **Details:**

This macro clears the Bit Error Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BEFClear(offset)`

### 3.5.2.30 Macro LINFLEX\_LLM\_BEFClear

LINCR1 registers - Bit Error Flag.

**Details:**

This macro read the bit indicating to the software that LINFlex has detected a bit error.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BEFClear(offset)`

### 3.5.2.31 Macro LINFLEX\_LLM\_BF\_Disable

LINCR1 registers - Bypass filter.

**Details:**

No interrupt if ID does not match any filter.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BF_Disable(offset)`

### 3.5.2.32 Macro LINFLEX\_LLM\_BF\_Enable

LINCR1 registers - Bypass filter.

**Details:**

An RX interrupt is generated on ID not matching any filter.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BF_Enable(offset)`

### 3.5.2.33 Macro LINFLEX\_LLM\_BOF\_Clear

LINESR registers - Buffer Overrun Flag.

#### Details:

This macro clears the Buffer Overrun Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BOF_Clear(offset)`

### 3.5.2.34 Macro LINFLEX\_LLM\_BOF\_Read

LINESR registers - Buffer Overrun Flag.

#### Details:

This macro read the bit indicating to the software that a new data byte is received and the buffer full flag is not cleared.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_BOF_Read(offset)`

### 3.5.2.35 Macro LINFLEX\_LLM\_CCD\_Disable

LINCR1 registers - Checksum Calculation Disable.

#### Details:

Checksum calculation is done by hardware.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CCD_Disable(offset)`

### 3.5.2.36 Macro LINFLEX\_LLM\_CCD\_Enable

LINCR1 registers - Checksum Calculation Enable.

**Details:**

Checksum calculation is done by software.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CCD_Enable(offset)`

### **3.5.2.37 Macro LINFLEX\_LLM\_CEF\_Clear**

LINESR registers - Checksum Error Flag.

**Details:**

This macro clears the Checksum Error Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CEF_Clear(offset)`

### **3.5.2.38 Macro LINFLEX\_LLM\_CEF\_Read**

LINESR registers - Checksum Error Flag.

**Details:**

This macro read the bit indicating to the software that the received checksum does not match the hardware calculated checksum.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CEF_Read(offset)`

### **3.5.2.39 Macro LINFLEX\_LLM\_CF\_Read**

LINCFR registers - Read Checksum bits.

**Details:**

Read the Checksum bits.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CF_Read(offset)`

### 3.5.2.40 Macro LINFLEX\_LLM\_CF\_Set

LINCFR registers - Set Header timeout value.

**Details:**

Set the Checksum bits.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_CF_Set(offset,value)`

### 3.5.2.41 Macro LINFLEX\_LLM\_CFD\_Disable

LINCR1 registers - Checksum field Disable.

**Details:**

Checksum field is sent after the required number of data bytes is sent.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CFD_Disable(offset)`

### 3.5.2.42 Macro LINFLEX\_LLM\_CFD\_Enable

LINCR1 registers - Checksum field Enable.

**Details:**

No checksum field is sent.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CFD_Enable(offset)`

### 3.5.2.43 Macro LINFLEX\_LLM\_CLASSIC\_CKS\_Set

BIDR registers - Set Classic Checksum for the frame.

**Details:**

Set Classic Checksum covering Data fields only. This is compatible with LIN specification 1.3 and below.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_CLASSIC_CKS_Set(offset)`

### 3.5.2.44 Macro LINFLEX\_LLM\_COUNTER\_VALUE\_Read

LINTCSR registers - Timeout counter enable.

**Details:**

Timeout counter enable. OCF bit is set if an output compare event occurs.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_COUNTER_VALUE_Read(offset)`

### 3.5.2.45 Macro LINFLEX\_LLM\_COUNTER\_VALUE\_Reset

LINTCSR registers - Timeout counter enable.

**Details:**

Timeout counter enable. OCF bit is set if an output compare event occurs.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_COUNTER_VALUE_Reset(offset)`

### 3.5.2.46 Macro LINFLEX\_LLM\_DATA0\_Clear

BDRL registers - Clear Data Byte 0.

#### Details:

Clear the Data byte 0 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA0\_Clear(offset)

### 3.5.2.47 Macro LINFLEX\_LLM\_DATA0\_Read

BDRL registers - Read Data Byte 0.

#### Details:

Read the Data byte 0 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA0\_Read(offset)

### 3.5.2.48 Macro LINFLEX\_LLM\_DATA0\_Set

BIDR registers - Set Data Byte 0.

#### Details:

Set the Data byte 0 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA0\_Set(offset,value)

### 3.5.2.49 Macro LINFLEX\_LLM\_DATA1\_Clear

BDRL registers - Clear Data Byte 1.

**Details:**

Clear the Data byte 1 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA1\_Clear(offset)

### 3.5.2.50 Macro LINFLEX\_LLM\_DATA1\_Read

BDRL registers - Read Data Byte 1.

**Details:**

Read the Data byte 1 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA1\_Read(offset)

### 3.5.2.51 Macro LINFLEX\_LLM\_DATA1\_Set

BIDR registers - Set Data Byte 1.

**Details:**

Set the Data byte 1 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA1\_Set(offset,value)



### 3.5.2.52 Macro LINFLEX\_LLM\_DATA2\_Clear

BDRL registers - Clear Data Byte 2.

#### Details:

Clear the Data byte 2 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA2\_Clear(offset)

### 3.5.2.53 Macro LINFLEX\_LLM\_DATA2\_Read

BDRL registers - Read Data Byte 2.

#### Details:

Read the Data byte 2 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA2\_Read(offset)

### 3.5.2.54 Macro LINFLEX\_LLM\_DATA2\_Set

BIDR registers - Set Data Byte 2.

#### Details:

Set the Data byte 2 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA2\_Set(offset,value)

### 3.5.2.55 Macro LINFLEX\_LLM\_DATA3\_Clear

BDRL registers - Clear Data Byte 3.

**Details:**

Clear the Data byte 3 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA3\_Clear(offset)

### 3.5.2.56 Macro LINFLEX\_LLM\_DATA3\_Read

BDRL registers - Read Data Byte 3.

**Details:**

Read the Data byte 3 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA3\_Read(offset)

### 3.5.2.57 Macro LINFLEX\_LLM\_DATA3\_Set

BIDR registers - Set Data Byte 3.

**Details:**

Set the Data byte 3 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA3\_Set(offset,value)

### 3.5.2.58 Macro LINFLEX\_LLM\_DATA4\_Clear

BDRL registers - Clear Data Byte 4.

#### Details:

Clear the Data byte 4 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA4\_Clear(offset)

### 3.5.2.59 Macro LINFLEX\_LLM\_DATA4\_Read

BDRL registers - Read Data Byte 4.

#### Details:

Read the Data byte 4 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA4\_Read(offset)

### 3.5.2.60 Macro LINFLEX\_LLM\_DATA4\_Set

BIDR registers - Set Data Byte 4.

#### Details:

Set the Data byte 4 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA4\_Set(offset,value)

### 3.5.2.61 Macro LINFLEX\_LLM\_DATA5\_Clear

BDRL registers - Clear Data Byte 5.

**Details:**

Clear the Data byte 5 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA5\_Clear(offset)

### 3.5.2.62 Macro LINFLEX\_LLM\_DATA5\_Read

BDRL registers - Read Data Byte 5.

**Details:**

Read the Data byte 5 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA5\_Read(offset)

### 3.5.2.63 Macro LINFLEX\_LLM\_DATA5\_Set

BIDR registers - Set Data Byte 5.

**Details:**

Set the Data byte 5 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA5\_Set(offset,value)

### 3.5.2.64 Macro LINFLEX\_LLM\_DATA6\_Clear

BDRL registers - Clear Data Byte 6.

#### Details:

Clear the Data byte 6 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA6\_Clear(offset)

### 3.5.2.65 Macro LINFLEX\_LLM\_DATA6\_Read

BDRL registers - Read Data Byte 6.

#### Details:

Read the Data byte 6 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA6\_Read(offset)

### 3.5.2.66 Macro LINFLEX\_LLM\_DATA6\_Set

BIDR registers - Set Data Byte 6.

#### Details:

Set the Data byte 6 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA6\_Set(offset,value)

### 3.5.2.67 Macro LINFLEX\_LLM\_DATA7\_Clear

BDRL registers - Clear Data Byte 7.

**Details:**

Clear the Data byte 7 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA7\_Clear(offset)

### 3.5.2.68 Macro LINFLEX\_LLM\_DATA7\_Read

BDRL registers - Read Data Byte 7.

**Details:**

Read the Data byte 7 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DATA7\_Read(offset)

### 3.5.2.69 Macro LINFLEX\_LLM\_DATA7\_Set

BIDR registers - Set Data Byte 7.

**Details:**

Set the Data byte 7 of the data field.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_DATA7\_Set(offset,value)

### 3.5.2.70 Macro LINFLEX\_LLM\_DATA\_BUFF\_EMPTY\_Clear

LINSR registers - Data Buffer Empty Flag.

#### Details:

This macro clears the Data Buffer Empty Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_DATA\_BUFF\_EMPTY\_Clear(offset)

### 3.5.2.71 Macro LINFLEX\_LLM\_DATA\_BUFF\_FULL\_Clear

LINSR registers - Data Buffer Full Flag.

#### Details:

This macro clears the Data Buffer Full Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_DATA\_BUFF\_FULL\_Clear(offset)

### 3.5.2.72 Macro LINFLEX\_LLM\_DATA\_RX\_COMPLETED\_Clear

LINSR registers - Data Reception Completed Flag.

#### Details:

This macro clears the Data Reception Completed Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_DATA\_RX\_COMPLETED\_Clear(offset)

### 3.5.2.73 Macro LINFLEX\_LLM\_DATA\_TX\_COMPLETED\_Clear

LINSR registers - Data Transmission Completed Flag.

#### Details:

This macro clears the Data Transmission Completed Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_DATA\_TX\_COMPLETED\_Clear(offset)

### 3.5.2.74 Macro LINFLEX\_LLM\_DDRQ\_Set

LINCR2 registers - Data Discard Request.

#### Details:

Setting this bit stop data reception if the frame does not concern the node.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DDRQ\_Set(offset)

### 3.5.2.75 Macro LINFLEX\_LLM\_DFL\_Clear

BIDR registers - Clear Data Field Length.

#### Details:

Clear the Data Field Length bits.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DFL\_Clear(offset)



### 3.5.2.76 Macro LINFLEX\_LLM\_DFL\_Extended\_Read

BIDR registers - Read Data Field Length for Extended frames.

**Details:**

Read DFL[0:5] where DFL[0:5] = Number of data bytes in the response part of the frame - 1.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DFL\_Extended\_Read(offset)

### 3.5.2.77 Macro LINFLEX\_LLM\_DFL\_Normal\_Read

BIDR registers - Read Data Field Length for Normal frames.

**Details:**

Read DFL[0:2] where DFL[0:2] = Number of data bytes in the response part of the frame - 1.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DFL\_Normal\_Read(offset)

### 3.5.2.78 Macro LINFLEX\_LLM\_DFL\_Set

BIDR registers - Set Data Field Length.

**Details:**

Set the Data Field Length.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** LINFLEX\_LLM\_DFL\_Set(offset,value)

### 3.5.2.79 Macro LINFLEX\_LLM\_DTRQ\_Set

LINCR2 registers - Data Transmission Request.

#### Details:

Set this bit in Slave mode to request the transmission of the LIN Data.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_DTRQ\_Set(offset)

### 3.5.2.80 Macro LINFLEX\_LLM\_ENHANCED\_CKS\_Set

BIDR registers - Set Enhanced Checksum for the frame.

#### Details:

Set Enhanced Checksum covering Identifier and Data fields. This is compatible with LIN specification 2.0 and higher.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_ENHANCED\_CKS\_Set(offset)

### 3.5.2.81 Macro LINFLEX\_LLM\_ENTER\_INIT\_MODE

LINCR1 registers - Enters Init Mode.

#### Details:

Enters the Init mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_ENTER\_INIT\_MODE(offset)

### 3.5.2.82 Macro LINFLEX\_LLM\_ENTER\_SLEEP\_MODE

LINCR1 registers - Enters Sleep Mode.

**Details:**

Enters the Sleep mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_ENTER\_SLEEP\_MODE(offset)

### 3.5.2.83 Macro LINFLEX\_LLM\_EXIT\_INIT\_MODE

LINCR1 registers - Exit Init Mode.

**Details:**

Exit the Init mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_EXIT\_INIT\_MODE(offset)

### 3.5.2.84 Macro LINFLEX\_LLM\_EXIT\_SLEEP\_MODE

LINCR1 registers - Exit Sleep Mode.

**Details:**

Exit the Sleep mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_EXIT\_SLEEP\_MODE(offset)

### 3.5.2.85 Macro LINFLEX\_LLM\_FEF\_Clear

LINESR registers - Framing Error Flag.

**Details:**

This macro clears the Framing Error Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_FEF_Clear(offset)`

### 3.5.2.86 Macro LINFLEX\_LLM\_FEF\_Read

LINESR registers - Framing Error Flag.

**Details:**

This macro read the bit indicating to the software that LINFlex has detected a framing error (invalid stop bit).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_FEF_Read(offset)`

### 3.5.2.87 Macro LINFLEX\_LLM\_HEADER\_RECEIVED

LINSR registers - Header Reception Flag.

**Details:**

This macro read the bit indicating to the software that valid header reception is completed.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_HEADER_RECEIVED(offset)`

### 3.5.2.88 Macro LINFLEX\_LLM\_HEADER\_RECEIVED\_Clear

LINSR registers - Header Reception Flag.

**Details:**

This macro clears the Header Reception Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_HEADER_RECEIVED_Clear(offset)`

### 3.5.2.89 Macro LINFLEX\_LLM\_HTO\_Read

LINTOCR registers - Read Header timeout value.

**Details:**

Read the Header timeout value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_HTO_Read(offset)`

### 3.5.2.90 Macro LINFLEX\_LLM\_HTO\_Set

LINTOCR registers - Set Header timeout value.

**Details:**

Set the Header timeout value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_HTO_Set(offset,value)`

### 3.5.2.91 Macro LINFLEX\_LLM\_HTRQ\_Set

LINCR2 registers - Set Header Transmission Request.

**Details:**

Request the transmission of the LIN header.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_HTRQ_Set(offset)`

### 3.5.2.92 Macro LINFLEX\_LLM\_ID\_Clear

BIDR registers - Clear Frame Identifier.

**Details:**

Clear the Identifier part of the identifier field without the identifier parity.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_ID_Clear(offset)`

### 3.5.2.93 Macro LINFLEX\_LLM\_ID\_Read

BIDR registers - Read Frame Identifier.

**Details:**

Read the Identifier part of the identifier field without the identifier parity.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_ID_Read(offset)`

### 3.5.2.94 Macro LINFLEX\_LLM\_ID\_Set

BIDR registers - Set Frame Identifier.

**Details:**

Set the Identifier part of the identifier field without the identifier parity.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_ID_Set(offset,value)`

### 3.5.2.95 Macro LINFLEX\_LLM\_IDPEF\_Clear

LINESR registers - Identifier Parity Error Flag.

**Details:**

This macro clears the Identifier Parity Error Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_IDPEF_Clear(offset)`

### 3.5.2.96 Macro LINFLEX\_LLM\_IDPEF\_Read

LINESR registers - Identifier Parity Error Flag.

**Details:**

This macro read the bit indicating to the software that an Identifier Parity error occurred.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_IDPEF_Read(offset)`

### 3.5.2.97 Macro LINFLEX\_LLM\_IS\_BDRL\_EMPTY

BDRL registers - Is the BDRL register empty ?

**Details:**

Verify if the BDRL register is empty.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_IS_BDRL_EMPTY(offset)`

### 3.5.2.98 Macro LINFLEX\_LLM\_IS\_BDRM\_EMPTY

BDRM registers - Is the BDRM register empty ?

**Details:**

Verify if the BDRM register is empty.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** LINFLEX\_LLM\_IS\_BDRM\_EMPTY(offset)

### 3.5.2.99 Macro LINFLEX\_LLM\_IS\_DATA\_BUFF\_EMPTY

LINSR registers - Data Buffer Empty Flag.

**Details:**

This macro read the bit indicating to the software that the buffer is empty.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_IS\_DATA\_BUFF\_EMPTY(offset)

### 3.5.2.100 Macro LINFLEX\_LLM\_IS\_DATA\_BUFF\_FULL

LINSR registers - Data Buffer Full Flag.

**Details:**

This macro read the bit indicating to the software that the buffer is full.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_IS\_DATA\_BUFF\_FULL(offset)



### 3.5.2.101 Macro LINFLEX\_LLM\_IS\_DATA\_RX\_COMPLETED

LINSR registers - Data Reception Completed Flag.

**Details:**

This macro read the bit indicating to the software that the data reception is completed.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_IS\_DATA\_RX\_COMPLETED(offset)

### 3.5.2.102 Macro LINFLEX\_LLM\_IS\_DATA\_TX\_COMPLETED

LINSR registers - Data Transmission Completed Flag.

**Details:**

This macro read the bit indicating to the software that the data TX is completed.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_IS\_DATA\_TX\_COMPLETED(offset)

### 3.5.2.103 Macro LINFLEX\_LLM\_IS\_MSG\_BUFFER\_FREE

LINSR registers - Release Message Buffer.

**Details:**

This macro read the Message Buffer to verify if it's free.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_IS\_MSG\_BUFFER\_FREE(offset)

### 3.5.2.104 Macro LINFLEX\_LLM\_IS\_RX\_BUSY

LINSR registers - Receiver Busy Flag.

#### Details:

This macro read the Receiver Busy Flag to verify if the receiver is idle or ongoing.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_IS\_RX\_BUSY(offset)

### 3.5.2.105 Macro LINFLEX\_LLM\_LIN\_BREAK\_10\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 10.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_10\_BIT\_Enable(offset)

### 3.5.2.106 Macro LINFLEX\_LLM\_LIN\_BREAK\_11\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 11.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_11\_BIT\_Enable(offset)

### 3.5.2.107 Macro LINFLEX\_LLM\_LIN\_BREAK\_12\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

**Details:**

The following macro set the Break length in Master mode to 12.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_12\_BIT\_Enable(offset)

### 3.5.2.108 Macro LINFLEX\_LLM\_LIN\_BREAK\_13\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

**Details:**

The following macro set the Break length in Master mode to 13.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_13\_BIT\_Enable(offset)

### 3.5.2.109 Macro LINFLEX\_LLM\_LIN\_BREAK\_14\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

**Details:**

The following macro set the Break length in Master mode to 14.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_14\_BIT\_Enable(offset)

### 3.5.2.110 Macro LINFLEX\_LLM\_LIN\_BREAK\_15\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 15.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_15\_BIT\_Enable(offset)

### 3.5.2.111 Macro LINFLEX\_LLM\_LIN\_BREAK\_16\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 16.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_16\_BIT\_Enable(offset)

### 3.5.2.112 Macro LINFLEX\_LLM\_LIN\_BREAK\_17\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 17.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_17\_BIT\_Enable(offset)

### 3.5.2.113 Macro LINFLEX\_LLM\_LIN\_BREAK\_18\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 18.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_18\_BIT\_Enable(offset)

### 3.5.2.114 Macro LINFLEX\_LLM\_LIN\_BREAK\_19\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 19.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_19\_BIT\_Enable(offset)

### 3.5.2.115 Macro LINFLEX\_LLM\_LIN\_BREAK\_20\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 20.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_20\_BIT\_Enable(offset)

**3.5.2.116 Macro LINFLEX\_LLM\_LIN\_BREAK\_21\_BIT\_Enable**

LINCR1 registers - LIN Master Break Length.

**Details:**

The following macro set the Break length in Master mode to 21.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_21\_BIT\_Enable(offset)

**3.5.2.117 Macro LINFLEX\_LLM\_LIN\_BREAK\_22\_BIT\_Enable**

LINCR1 registers - LIN Master Break Length.

**Details:**

The following macro set the Break length in Master mode to 22.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_22\_BIT\_Enable(offset)

**3.5.2.118 Macro LINFLEX\_LLM\_LIN\_BREAK\_23\_BIT\_Enable**

LINCR1 registers - LIN Master Break Length.

**Details:**

The following macro set the Break length in Master mode to 23.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_23\_BIT\_Enable(offset)

### 3.5.2.119 Macro LINFLEX\_LLM\_LIN\_BREAK\_36\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 36.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_36\_BIT\_Enable(offset)

### 3.5.2.120 Macro LINFLEX\_LLM\_LIN\_BREAK\_50\_BIT\_Enable

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to 50.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_BREAK\_50\_BIT\_Enable(offset)

### 3.5.2.121 Macro LINFLEX\_LLM\_LIN\_BREAK\_LENGTH\_Set

LINCR1 registers - LIN Master Break Length.

#### Details:

The following macro set the Break length in Master mode to the value parameter.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_LIN_BREAK_LENGTH_Set(offset,value)`

### 3.5.2.122 Macro LINFLEX\_LLM\_LIN\_IRQ\_BEIE\_Disable

LINIER registers - Bit Error Interrupt Enable.

**Details:**

This macro disables the LINESR[BEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_BEIE_Disable(offset)`

### 3.5.2.123 Macro LINFLEX\_LLM\_LIN\_IRQ\_BEIE\_Enable

LINIER registers - Bit Error Interrupt Enable.

**Details:**

This macro enables the LINESR[BEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_BEIE_Enable(offset)`

### 3.5.2.124 Macro LINFLEX\_LLM\_LIN\_IRQ\_BEIE\_Read

LINIER registers - Bit Error Interrupt Enable.

**Details:**

This macro read the LINESR[BEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.



**Prototype:** `LINFLEX_LLM_LIN_IRQ_BEIE_Read(offset)`

### 3.5.2.125 Macro LINFLEX\_LLM\_LIN\_IRQ\_BOIE\_Disable

LINIER registers - Buffer Overrun Interrupt Enable.

**Details:**

This macro disables the LINESR[BOF] or UARTSR[BOF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_BOIE_Disable(offset)`

### 3.5.2.126 Macro LINFLEX\_LLM\_LIN\_IRQ\_BOIE\_Enable

LINIER registers - Buffer Overrun Interrupt Enable.

**Details:**

This macro enables the LINESR[BOF] or UARTSR[BOF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_BOIE_Enable(offset)`

### 3.5.2.127 Macro LINFLEX\_LLM\_LIN\_IRQ\_BOIE\_Read

LINIER registers - Buffer Overrun Interrupt Enable.

**Details:**

This macro read the LINESR[BOF] or UARTSR[BOF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_BOIE_Read(offset)`

### 3.5.2.128 Macro LINFLEX\_LLM\_LIN\_IRQ\_CEIE\_Disable

LINIER registers - Checksum Error Interrupt Enable.

**Details:**

This macro disables the LINESR[CEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_CEIE\_Disable(offset)

### 3.5.2.129 Macro LINFLEX\_LLM\_LIN\_IRQ\_CEIE\_Enable

LINIER registers - Checksum Error Interrupt Enable.

**Details:**

This macro enables the LINESR[CEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_CEIE\_Enable(offset)

### 3.5.2.130 Macro LINFLEX\_LLM\_LIN\_IRQ\_CEIE\_Read

LINIER registers - Checksum Error Interrupt Enable.

**Details:**

This macro read the LINESR[CEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_CEIE\_Read(offset)

### 3.5.2.131 Macro LINFLEX\_LLM\_LIN\_IRQ\_DBEIE\_Disable

LINIER registers - Data Buffer Empty Interrupt Enable.

**Details:**

This macro disables the interrupt request generation when data buffer register is empty.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DBEIE\_Disable(offset)

### 3.5.2.132 Macro LINFLEX\_LLM\_LIN\_IRQ\_DBEIE\_Enable

LINIER registers - Data Buffer Empty Interrupt Enable.

**Details:**

This macro enables the interrupt request generation when data buffer register is empty.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DBEIE\_Enable(offset)

### 3.5.2.133 Macro LINFLEX\_LLM\_LIN\_IRQ\_DBEIE\_Read

LINIER registers - Data Buffer Empty Interrupt Enable.

**Details:**

This macro read the interrupt request generation when data buffer register is empty.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DBEIE\_Read(offset)

### 3.5.2.134 Macro LINFLEX\_LLM\_LIN\_IRQ\_DBFIE\_Disable

LINIER registers - Data Buffer Full Interrupt Enable.

#### Details:

This macro disables the interrupt request generation when data buffer register is full.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DBFIE\_Disable(offset)

### 3.5.2.135 Macro LINFLEX\_LLM\_LIN\_IRQ\_DBFIE\_Enable

LINIER registers - Data Buffer Full Interrupt Enable.

#### Details:

This macro enables the interrupt request generation when data buffer register is full.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DBFIE\_Enable(offset)

### 3.5.2.136 Macro LINFLEX\_LLM\_LIN\_IRQ\_DBFIE\_Read

LINIER registers - Data Buffer Full Interrupt Enable.

#### Details:

This macro read the interrupt request generation when data buffer register is full.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DBFIE\_Read(offset)

### 3.5.2.137 Macro LINFLEX\_LLM\_LIN\_IRQ\_DRIE\_Disable

LINIER registers - Data Reception Complete Interrupt Enable.

#### Details:

This macro disables the LINSR[DRF] or UARTSR[DRF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DRIE\_Disable(offset)

### 3.5.2.138 Macro LINFLEX\_LLM\_LIN\_IRQ\_DRIE\_Enable

LINIER registers - Data Reception Complete Interrupt Enable.

#### Details:

This macro enables the LINSR[DRF] or UARTSR[DRF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DRIE\_Enable(offset)

### 3.5.2.139 Macro LINFLEX\_LLM\_LIN\_IRQ\_DRIE\_Read

LINIER registers - Data Reception Complete Interrupt Enable.

#### Details:

This macro read the LINSR[DRF] or UARTSR[DRF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DRIE\_Read(offset)

### 3.5.2.140 Macro LINFLEX\_LLM\_LIN\_IRQ\_DTIE\_Disable

LINIER registers - Data Transmitted Complete Interrupt Enable.

**Details:**

This macro disables the LINSR[DTF] or UARTSR[DTF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DTIE\_Disable(offset)

**3.5.2.141 Macro LINFLEX\_LLM\_LIN\_IRQ\_DTIE\_Enable**

LINIER registers - Data Transmitted Complete Interrupt Enable.

**Details:**

This macro enables the LINSR[DTF] or UARTSR[DTF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DTIE\_Enable(offset)

**3.5.2.142 Macro LINFLEX\_LLM\_LIN\_IRQ\_DTIE\_Read**

LINIER registers - Data Transmitted Complete Interrupt Enable.

**Details:**

This macro read the LINSR[DTF] or UARTSR[DTF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_DTIE\_Read(offset)

**3.5.2.143 Macro LINFLEX\_LLM\_LIN\_IRQ\_FEIE\_Disable**

LINIER registers - Framing Error Interrupt Enable.

**Details:**

This macro disables the LINESR[FEF] or UARTSR[FEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_FEIE\_Disable(offset)

### 3.5.2.144 Macro LINFLEX\_LLM\_LIN\_IRQ\_FEIE\_Enable

LINIER registers - Framing Error Interrupt Enable.

**Details:**

This macro enables the LINESR[FEF] or UARTSR[FEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_FEIE\_Enable(offset)

### 3.5.2.145 Macro LINFLEX\_LLM\_LIN\_IRQ\_FEIE\_Read

LINIER registers - Framing Error Interrupt Enable.

**Details:**

This macro read the LINESR[FEF] or UARTSR[FEF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_FEIE\_Read(offset)

### 3.5.2.146 Macro LINFLEX\_LLM\_LIN\_IRQ\_HEIE\_Disable

LINIER registers - Header Error Interrupt Enable.

**Details:**

This macro disables LINESR[BDEF], LINESR[SFEF], LINESR[IDPEF] interrupt requests generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_HEIE_Disable(offset)`

### 3.5.2.147 Macro LINFLEX\_LLM\_LIN\_IRQ\_HEIE\_Enable

LINIER registers - Header Error Interrupt Enable.

**Details:**

This macro enables the LINESR[BDEF], LINESR[SFEF], LINESR[IDPEF] interrupt requests generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_HEIE_Enable(offset)`

### 3.5.2.148 Macro LINFLEX\_LLM\_LIN\_IRQ\_HEIE\_Read

LINIER registers - Header Error Interrupt Enable.

**Details:**

This macro read the LINESR[BDEF], LINESR[SFEF], LINESR[IDPEF] interrupt requests generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_HEIE_Read(offset)`

### 3.5.2.149 Macro LINFLEX\_LLM\_LIN\_IRQ\_HRIE\_Disable

LINIER registers - Header Received Interrupt Enable.

**Details:**

This macro disables the LINSR[HRF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.



**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_HRIE_Disable(offset)`

### 3.5.2.150 Macro LINFLEX\_LLM\_LIN\_IRQ\_HRIE\_Enable

LINIER registers - Header Received Interrupt Enable.

**Details:**

This macro enables the LINSR[HRF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_HRIE_Enable(offset)`

### 3.5.2.151 Macro LINFLEX\_LLM\_LIN\_IRQ\_HRIE\_Read

LINIER registers - Header Received Interrupt Enable.

**Details:**

This macro read the LINSR[HRF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_HRIE_Read(offset)`

### 3.5.2.152 Macro LINFLEX\_LLM\_LIN\_IRQ\_LSIE\_Disable

LINIER registers - LIN State Interrupt Enable.

**Details:**

This macro disables the LIN state change (LINSR) interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_LSIE_Disable(offset)`

**3.5.2.153 Macro LINFLEX\_LLM\_LIN\_IRQ\_LSIE\_Enable**

LINIER registers - LIN State Interrupt Enable.

**Details:**

This macro enables the LIN state change (LINSR) interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_LSIE\_Enable(offset)

**3.5.2.154 Macro LINFLEX\_LLM\_LIN\_IRQ\_LSIE\_Read**

LINIER registers - LIN State Interrupt Enable.

**Details:**

This macro read the LIN state change (LINSR) interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_LSIE\_Read(offset)

**3.5.2.155 Macro LINFLEX\_LLM\_LIN\_IRQ\_OCIE\_Disable**

LINIER registers - Output Compare Interrupt Enable.

**Details:**

This macro disables the LINESR[OCF] or UARTSR[OCF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_OCIE\_Disable(offset)

### 3.5.2.156 Macro LINFLEX\_LLM\_LIN\_IRQ\_OCIE\_Enable

LINIER registers - Output Compare Interrupt Enable.

#### Details:

This macro enables the LINESR[OCF] or UARTSR[OCF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_OCIE\_Enable(offset)

### 3.5.2.157 Macro LINFLEX\_LLM\_LIN\_IRQ\_OCIE\_Read

LINIER registers - Output Compare Interrupt Enable.

#### Details:

This macro read the LINESR[OCF] or UARTSR[OCF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_OCIE\_Read(offset)

### 3.5.2.158 Macro LINFLEX\_LLM\_LIN\_IRQ\_SZIE\_Disable

LINIER registers - Stuck at Zero Interrupt Disable.

#### Details:

This macro disables the LINESR[SZF] or UARTSR[SZF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_SZIE\_Disable(offset)

### 3.5.2.159 Macro LINFLEX\_LLM\_LIN\_IRQ\_SZIE\_Enable

LINIER registers - Stuck at Zero Interrupt Enable.

**Details:**

This macro enables the LINESR[SZF] or UARTSR[SZF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_SZIE\_Enable(offset)

### **3.5.2.160 Macro LINFLEX\_LLM\_LIN\_IRQ\_SZIE\_Read**

LINIER registers - Stuck at Zero Interrupt Read.

**Details:**

This macro read the LINESR[SZF] or UARTSR[SZF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_SZIE\_Read(offset)

### **3.5.2.161 Macro LINFLEX\_LLM\_LIN\_IRQ\_WUIE\_Disable**

LINIER registers - Wake-up Interrupt Enable.

**Details:**

This macro disables the LINSR[WUF] or UARTSR[WUF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_IRQ\_WUIE\_Disable(offset)

### **3.5.2.162 Macro LINFLEX\_LLM\_LIN\_IRQ\_WUIE\_Enable**

LINIER registers - Wake-up Interrupt Enable.

**Details:**

This macro enables the LINSR[WUF] or UARTSR[WUF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_WUIE_Enable(offset)`

### 3.5.2.163 Macro LINFLEX\_LLM\_LIN\_IRQ\_WUIE\_Read

LINIER registers - Wake-up Interrupt Enable.

**Details:**

This macro read the LINSR[WUF] or UARTSR[WUF] interrupt request generation.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_IRQ_WUIE_Read(offset)`

### 3.5.2.164 Macro LINFLEX\_LLM\_LIN\_MODE\_Set

UARTCR registers - LIN mode enable.

**Details:**

Enables the LIN mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LIN_MODE_Set(offset)`

### 3.5.2.165 Macro LINFLEX\_LLM\_LIN\_STATE\_Read

LINSR registers - LIN state.

**Details:**

This macro returns the LIN State: 0 Sleep mode: LINFlex is in Sleep mode to save power consumption 1 Initialization mode: LINFlex is in Initialization mode 2 Idle mode: This state is entered on several events: SLEEP bit and INIT in LINCR1 register have been cleared by software, A falling edge has been received on RX pin and AWUM bit is set,

The previous frame reception or transmission has been completed or aborted 3 Break transmission ongoing 4 Break transmission has been completed. Break Delimiter transmission is ongoing 5 Synch Field transmission is ongoing 6 Identifier transmission is ongoing 7 Header transmission is completed 8 Response reception/transmission is ongoing 9 Data reception/transmission completed. Checksum RX/TX ongoing F Sleep mode: Reset LIN State Interrupt (generated on LIN state change)

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LIN\_STATE\_Read(offset)

### 3.5.2.166 Macro LINFLEX\_LLM\_LIN\_TIMEOUT\_MODE\_Set

LINTCSR registers - Output compare mode.

**Details:**

Set the Output compare mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_LIN\_TIMEOUT\_MODE\_Set(offset)

### 3.5.2.167 Macro LINFLEX\_LLM\_LINCR1\_Reset

LINFLEX registers - LINCR1 register bit masks.

**Details:**

Reset LINCR1 macro.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_LINCR1\_Reset(offset)

### 3.5.2.168 Macro LINFLEX\_LLM\_LINTCSR\_Reset

LINFLEX registers - LINTCSR register bit masks.

**Details:**

Reset LINTCSR macro.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_LINTCSR_Reset(offset)`

### 3.5.2.169 Macro LINFLEX\_LLM\_LOOP\_BACK\_MODE\_Enable

LINCR1 registers - Loop Back mode enable.

**Details:**

Enable the Loop Back mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_LOOP_BACK_MODE_Enable(offset)`

### 3.5.2.170 Macro LINFLEX\_LLM\_MASTER\_Enable

LINCR1 registers - Master Mode Enable.

**Details:**

Enable the Master mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_MASTER_Enable(offset)`

### 3.5.2.171 Macro LINFLEX\_LLM\_NF\_Clear

LINESR registers - Noise Flag.

**Details:**

This macro clears the Noise Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_NF_Clear(offset)`

### 3.5.2.172 Macro **LINFLEX\_LLM\_NF\_Read**

LINESR registers - Noise Flag.

**Details:**

This macro read the bit indicating to the software that a noise is detected on a received character.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_NF_Read(offset)`

### 3.5.2.173 Macro **LINFLEX\_LLM\_NOT\_RESET\_STATE\_MACHINE\_ON\_IOBE\_Set**

LINCR2 registers - LIN state machine does not reset on Bit Error.

**Details:**

Bit error does not reset LIN state machine.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_NOT_RESET_STATE_MACHINE_ON_IOBE_Set(offset)`

### 3.5.2.174 Macro **LINFLEX\_LLM\_NOT\_RESET\_STATE\_MACHINE\_ON\_IOPE\_Set**

LINCR2 registers - LIN state machine does not reset on Identifier Parity Error.

**Details:**



Identifier Parity error does not reset LIN state machine.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_NOT_RESET_STATE_MACHINE_ON_IOPE_Set(offset)`

### 3.5.2.175 Macro

#### **LINFLEX\_LLM\_NOT\_RESET\_STATE\_MACHINE\_TO\_IOT\_Set**

LINTCSR registers - Idle on Timeout.

#### **Details:**

LIN state machine not reset to Idle on timeout event.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_NOT_RESET_STATE_MACHINE_TO_IOT_Set(offset)`

### 3.5.2.176 Macro LINFLEX\_LLM\_OC1\_Read

LINOCR registers - Read Output compare 1 value.

#### **Details:**

Read the Output compare 1 value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_OC1_Read(offset)`

### 3.5.2.177 Macro LINFLEX\_LLM\_OC1\_Set

LINOCR registers - Set Output compare 1 value.

**Details:**

Set the Output compare 1 value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_OC1_Set(offset,value)`

### **3.5.2.178 Macro LINFLEX\_LLM\_OC2\_Read**

LINOCR registers - Read Output compare 2 value.

**Details:**

Read the Output compare 2 value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_OC2_Read(offset)`

### **3.5.2.179 Macro LINFLEX\_LLM\_OC2\_Set**

LINOCR registers - Set Output compare 2 value.

**Details:**

Set the Output compare 2 value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_OC2_Set(offset,value)`

### **3.5.2.180 Macro LINFLEX\_LLM\_OCF\_Clear**

LINESR registers - Output Compare Flag.

**Details:**

This macro clears the Output Compare Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_OCF_Clear(offset)`

### **3.5.2.181 Macro LINFLEX\_LLM\_OCF\_Read**

LINESR registers - Output Compare Flag.

**Details:**

This macro read the OCF bit. If this bit is set and IOT bit in LINTCSR is set, LINFlex moves to Idle state.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_OCF_Read(offset)`

### **3.5.2.182 Macro LINFLEX\_LLM\_OUTPUT\_COMPARE\_MODE\_Set**

LINTCSR registers - LIN timeout mode.

**Details:**

Set the LIN timeout mode (header, response and frame timeout detection).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_OUTPUT_COMPARE_MODE_Set(offset)`

### **3.5.2.183 Macro LINFLEX\_LLM\_RELEASE\_MESSAGE\_BUFFER**

LINSR registers - Release Message Buffer.

**Details:**

This macro releases the Message Buffer.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_RELEASE_MESSAGE_BUFFER(offset)`

### 3.5.2.184 Macro LINFLEX\_LLM\_RESET\_BAUDRATE

LINFBRR/LINIBRR registers - Set baudrate macro.

#### **Details:**

It has a limitation for LINIBRR value on 16 bits only On Leopard this value is on 20 bits, but considering LIN 2.x standard and maximum clock rate for peripherals (120 MHz) this value cannot exceed 16 bits In case this limitation has to be avoided, also plugins modules have to be modified.

**Prototype:** `LINFLEX_LLM_RESET_BAUDRATE(offset)`

### 3.5.2.185 Macro LINFLEX\_LLM\_RESET\_STATE\_MACHINE\_ON\_IOBE\_Set

LINCR2 registers - LIN state machine reset on Bit Error.

#### **Details:**

Bit error reset LIN state machine.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_RESET_STATE_MACHINE_ON_IOBE_Set(offset)`

### 3.5.2.186 Macro **LINFLEX\_LLM\_RESET\_STATE\_MACHINE\_ON\_IOPE\_Set**

LINCR2 registers - LIN state machine reset on Identifier Parity Error.

#### Details:

Identifier Parity error reset LIN state machine.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_RESET_STATE_MACHINE_ON_IOPE_Set(offset)`

### 3.5.2.187 Macro **LINFLEX\_LLM\_RESET\_STATE\_MACHINE\_TO\_IOT\_Set**

LINTCSR registers - Idle on Timeout.

#### Details:

LIN state machine reset to Idle on timeout event.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_RESET_STATE_MACHINE_TO_IOT_Set(offset)`

### 3.5.2.188 Macro **LINFLEX\_LLM\_RTO\_Read**

LINTOCR registers - Read Response timeout value.

#### Details:

Read the Response timeout value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_RTO_Read(offset)`

### 3.5.2.189 Macro LINFLEX\_LLM\_RTO\_Set

LINTOCR registers - Set Response timeout value.

**Details:**

Set the Response timeout value.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Prototype:** `LINFLEX_LLM_RTO_Set(offset,value)`

### 3.5.2.190 Macro LINFLEX\_LLM\_RX\_BUFF\_1\_BYTE\_Set

UARTCR registers - Transmitter Data Field length.

**Details:**

The following macro set the number of bytes to be received in UART mode to 1.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_RX_BUFF_1_BYTE_Set(offset)`

### 3.5.2.191 Macro LINFLEX\_LLM\_RX\_BUFF\_2\_BYTES\_Set

UARTCR registers - Transmitter Data Field length.

**Details:**

The following macro set the number of bytes to be received in UART mode to 2.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_RX_BUFF_2_BYTES_Set(offset)`

**3.5.2.192 Macro LINFLEX\_LLM\_RX\_BUFF\_3\_BYTES\_Set**

UARTCR registers - Transmitter Data Field length.

**Details:**

The following macro set the number of bytes to be received in UART mode to 3.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_RX_BUFF_3_BYTES_Set(offset)`

**3.5.2.193 Macro LINFLEX\_LLM\_RX\_BUFF\_4\_BYTES\_Set**

UARTCR registers - Transmitter Data Field length.

**Details:**

The following macro set the number of bytes to be received in UART mode to 4.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_RX_BUFF_4_BYTES_Set(offset)`

**3.5.2.194 Macro LINFLEX\_LLM\_RX\_BUSY\_Clear**

LINSR registers - Receiver Busy Flag.

**Details:**

This macro clears the Receiver Busy Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_RX_BUSY_Clear(offset)`

**3.5.2.195 Macro LINFLEX\_LLM\_RX\_DIRECTION\_Set**

BIDR registers - Set Rx direction of the frame.

**Details:**

LINFLEX receives the data and copy them in the BDRL and BDRM registers.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_RX\_DIRECTION\_Set(offset)

### **3.5.2.196 Macro LINFLEX\_LLM\_RX\_PIN\_STATE\_Read**

LINSR registers - LIN receive pin state.

**Details:**

This macro read the bit reflecting the current status of LINRX pin for diagnostic purposes.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_RX\_PIN\_STATE\_Read(offset)

### **3.5.2.197 Macro LINFLEX\_LLM\_SELF\_TEST\_MODE\_Enable**

LINCR1 registers - Self Test mode enable.

**Details:**

Enable the Self Test mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Violates:** Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

**Violates:** C macros shall only expand to a braced initializer...

**Prototype:** LINFLEX\_LLM\_SELF\_TEST\_MODE\_Enable(offset)



### 3.5.2.198 Macro LINFLEX\_LLM\_SET\_BAUDRATE

**Prototype:** LINFLEX\_LLM\_SET\_BAUDRATE(offset,value)

### 3.5.2.199 Macro LINFLEX\_LLM\_SFEF\_Clear

LINESR registers - Synch Field Error Flag.

**Details:**

This macro clears the Synch Field Error Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_SFEF\_Clear(offset)

### 3.5.2.200 Macro LINFLEX\_LLM\_SFEF\_Read

LINESR registers - Synch Field Error Flag.

**Details:**

This macro read the bit indicating to the software that Synch Field error occurred (inconsistent Synch Field).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_SFEF\_Read(offset)

### 3.5.2.201 Macro LINFLEX\_LLM\_SLAVE\_Enable

LINCR1 registers - Slave Mode Enable.

**Details:**

Enable the Slave mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_SLAVE\_Enable(offset)

**3.5.2.202 Macro LINFLEX\_LLM\_SZF\_Clear**

LINESR registers - Stuck at zero Flag.

**Details:**

This macro clears the Stuck at zero Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_SZF\_Clear(offset)

**3.5.2.203 Macro LINFLEX\_LLM\_SZF\_Read**

LINESR registers - Stuck at zero Flag.

**Details:**

This macro read the bit indicating to the software that the bus is dominant for more than a 100-bit time.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_SZF\_Read(offset)

**3.5.2.204 Macro LINFLEX\_LLM\_TIMEOUT\_COUNTER\_Disable**

LINTCSR registers - Timeout counter disable.

**Details:**

Timeout counter disable. OCF bit in LINESR or UARTSR is not set on an output compare event

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_TIMEOUT\_COUNTER\_Disable(offset)

### 3.5.2.205 Macro LINFLEX\_LLM\_TIMEOUT\_COUNTER\_Enable

LINTCSR registers - Timeout counter enable.

#### Details:

Timeout counter enable. OCF bit is set if an output compare event occurs.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** LINFLEX\_LLM\_TIMEOUT\_COUNTER\_Enable(offset)

### 3.5.2.206 Macro LINFLEX\_LLM\_TX\_BUFF\_1\_BYTE\_Set

UARTCR registers - Transmitter Data Field length.

#### Details:

The following macro set the number of bytes to be transmitted in UART mode to 1.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_TX\_BUFF\_1\_BYTE\_Set(offset)

### 3.5.2.207 Macro LINFLEX\_LLM\_TX\_BUFF\_2\_BYTES\_Set

UARTCR registers - Transmitter Data Field length.

#### Details:

The following macro set the number of bytes to be transmitted in UART mode to 2.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_TX\_BUFF\_2\_BYTES\_Set(offset)

### 3.5.2.208 Macro LINFLEX\_LLM\_TX\_BUFF\_3\_BYTES\_Set

UARTCR registers - Transmitter Data Field length.

#### Details:

The following macro set the number of bytes to be transmitted in UART mode to 3.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_TX\_BUFF\_3\_BYTES\_Set(offset)

### 3.5.2.209 Macro LINFLEX\_LLM\_TX\_BUFF\_4\_BYTES\_Set

UARTCR registers - Transmitter Data Field length.

#### Details:

The following macro set the number of bytes to be transmitted in UART mode to 4.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_TX\_BUFF\_4\_BYTES\_Set(offset)

### 3.5.2.210 Macro LINFLEX\_LLM\_TX\_DIRECTION\_Set

BIDR registers - Set Tx direction of the frame.

#### Details:

LINFLEX transmits the data from the BDR registers.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_TX\_DIRECTION\_Set(offset)

### 3.5.2.211 Macro LINFLEX\_LLM\_UART\_BOF\_Clear

UARTSR registers - Buffer Overrun Flag.

#### Details:

This macro clears the Buffer Overrun Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_BOF_Clear(offset)`

### 3.5.2.212 Macro LINFLEX\_LLM\_UART\_BOF\_Read

UARTSR registers - Buffer Overrun Flag.

#### **Details:**

This macro read the bit indicating to the software that a new data byte is received and the buffer full flag is not cleared.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_BOF_Read(offset)`

### 3.5.2.213 Macro LINFLEX\_LLM\_UART\_DRF\_Clear

UARTSR registers - Data Reception Completed Flag.

#### **Details:**

This macro clears the Data Reception Completed Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_DRF_Clear(offset)`

### 3.5.2.214 Macro LINFLEX\_LLM\_UART\_DRF\_Read

UARTSR registers - Data Reception Completed Flag.

#### **Details:**

This macro read the bit indicating to the software that the data reception is completed.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_DRF_Read(offset)`

### 3.5.2.215 Macro LINFLEX\_LLM\_UART\_DTF\_Clear

UARTSR registers - Data Transmission Completed Flag.

**Details:**

This macro clears the Data Transmission Completed Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_DTF_Clear(offset)`

### 3.5.2.216 Macro LINFLEX\_LLM\_UART\_DTF\_Read

UARTSR registers - Data Transmission Completed Flag.

**Details:**

This macro read the bit indicating to the software that the data transmission is completed.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_DTF_Read(offset)`

### 3.5.2.217 Macro LINFLEX\_LLM\_UART\_EVEN\_PARITY\_Set

UARTCR registers - Even Parity.

**Details:**

Set the Even Parity in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_UART_EVEN_PARITY_Set(offset)`

### 3.5.2.218 Macro LINFLEX\_LLM\_UART\_FEF\_Clear

UARTSR registers - Framing Error Flag.

#### Details:

This macro clears the Framing Error Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_FEF\_Clear(offset)

### 3.5.2.219 Macro LINFLEX\_LLM\_UART\_FEF\_Read

UARTSR registers - Framing Error Flag.

#### Details:

This macro read the bit indicating to the software that LINFlex has detected a framing error (invalid stop bit).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_FEF\_Read(offset)

### 3.5.2.220 Macro LINFLEX\_LLM\_UART\_MODE\_Set

UARTCR registers - UART mode enable.

#### Details:

Enables the UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_MODE\_Set(offset)

### 3.5.2.221 Macro LINFLEX\_LLM\_UART\_NF\_Clear

UARTSR registers - Noise Flag.

#### Details:

This macro clears the Noise Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_NF\_Clear(offset)

### 3.5.2.222 Macro LINFLEX\_LLM\_UART\_NF\_Read

UARTSR registers - Noise Flag.

#### Details:

This macro read the bit indicating to the software that noise is detected on a received character.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_NF\_Read(offset)

### 3.5.2.223 Macro LINFLEX\_LLM\_UART\_OCF\_Clear

UARTSR registers - Output Compare Flag.

#### Details:

This macro clears the Output Compare Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_OCF\_Clear(offset)

### 3.5.2.224 Macro LINFLEX\_LLM\_UART\_OCF\_Read

UARTSR registers - Output Compare Flag.



**Details:**

This macro read the bit indicating to the software that the counter has matched the content of OC1[0:7] or OC2[0:7] in LINOCR.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_OCF_Read(offset)`

**3.5.2.225 Macro LINFLEX\_LLM\_UART\_ODD\_PARITY\_Set**

UARTCR registers - Odd Parity.

**Details:**

Set the Odd Parity in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_ODD_PARITY_Set(offset)`

**3.5.2.226 Macro LINFLEX\_LLM\_UART\_PARITY\_CONTROL\_Disable**

UARTCR registers - Parity Control Disable.

**Details:**

Disable the Parity transmit/check in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_UART_PARITY_CONTROL_Disable(offset)`

**3.5.2.227 Macro LINFLEX\_LLM\_UART\_PARITY\_CONTROL\_Enable**

UARTCR registers - Parity Control Enable.

**Details:**

Enable the Parity transmit/check in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Violates:** This is required as per Autosar Requirement.

**Prototype:** `LINFLEX_LLM_UART_PARITY_CONTROL_Enable(offset)`

### **3.5.2.228 Macro LINFLEX\_LLM\_UART\_PE0\_Clear**

UARTSR registers - Parity Error Flag Rx0.

**Details:**

This macro clears the Parity Error Flag Rx0.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE0_Clear(offset)`

### **3.5.2.229 Macro LINFLEX\_LLM\_UART\_PE0\_Read**

UARTSR registers - Parity Error Flag Rx0.

**Details:**

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx0).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE0_Read(offset)`

### **3.5.2.230 Macro LINFLEX\_LLM\_UART\_PE1\_Clear**

UARTSR registers - Parity Error Flag Rx1.

**Details:**

This macro clears the Parity Error Flag Rx1.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE1_Clear(offset)`

### 3.5.2.231 Macro LINFLEX\_LLM\_UART\_PE1\_Read

UARTSR registers - Parity Error Flag Rx1.

#### **Details:**

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx1).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE1_Read(offset)`

### 3.5.2.232 Macro LINFLEX\_LLM\_UART\_PE2\_Clear

UARTSR registers - Parity Error Flag Rx2.

#### **Details:**

This macro clears the Parity Error Flag Rx2.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE2_Clear(offset)`

### 3.5.2.233 Macro LINFLEX\_LLM\_UART\_PE2\_Read

UARTSR registers - Parity Error Flag Rx2.

#### **Details:**

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx2).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE2_Read(offset)`

### 3.5.2.234 Macro LINFLEX\_LLM\_UART\_PE3\_Clear

UARTSR registers - Parity Error Flag Rx3.

**Details:**

This macro clears the Parity Error Flag Rx3.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE3_Clear(offset)`

### 3.5.2.235 Macro LINFLEX\_LLM\_UART\_PE3\_Read

UARTSR registers - Parity Error Flag Rx3.

**Details:**

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx3).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_PE3_Read(offset)`

### 3.5.2.236 Macro LINFLEX\_LLM\_UART\_RMB\_Clear

UARTSR registers - Release Message Buffer.

**Details:**

This macro releases the Message Buffer.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_RMB_Clear(offset)`

### 3.5.2.237 Macro LINFLEX\_LLM\_UART\_RMB\_Read

UARTSR registers - Release Message Buffer.

#### Details:

This macro read the Message Buffer to verify if it's free.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_RMB\_Read(offset)

### 3.5.2.238 Macro LINFLEX\_LLM\_UART\_RPS\_Read

UARTSR registers - UART receive pin state.

#### Details:

This macro read the bit reflecting the current status of UARTRX pin for diagnostic purposes.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_RPS\_Read(offset)

### 3.5.2.239 Macro LINFLEX\_LLM\_UART\_RX\_Disable

UARTCR registers - Receiver Disable.

#### Details:

Disables the RX in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_RX\_Disable(offset)

### 3.5.2.240 Macro LINFLEX\_LLM\_UART\_RX\_Enable

UARTCR registers - Receiver Enable.

**Details:**

Enables the RX in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_RX\_Enable(offset)

### 3.5.2.241 Macro LINFLEX\_LLM\_UART\_SZF\_Clear

UARTSR registers - Stuck at zero Flag.

**Details:**

This macro clears the Stuck at zero Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_SZF\_Clear(offset)

### 3.5.2.242 Macro LINFLEX\_LLM\_UART\_SZF\_Read

UARTSR registers - Stuck at zero Flag.

**Details:**

This macro read the bit indicating to the software that bus is dominant for more than a 100-bit time.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** LINFLEX\_LLM\_UART\_SZF\_Read(offset)

### 3.5.2.243 Macro LINFLEX\_LLM\_UART\_TX\_Disable

UARTCR registers - Transmitter Disable.

**Details:**

Disables the TX in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_TX_Disable(offset)`

**3.5.2.244 Macro LINFLEX\_LLM\_UART\_TX\_Enable**

UARTCR registers - Transmitter Enable.

**Details:**

Enables the TX in UART mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_TX_Enable(offset)`

**3.5.2.245 Macro LINFLEX\_LLM\_UART\_WUF\_Clear**

UARTSR registers - Wake-up Flag.

**Details:**

This macro clears the Wake-up Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_WUF_Clear(offset)`

**3.5.2.246 Macro LINFLEX\_LLM\_UART\_WUF\_Read**

UARTSR registers - Wake-up Flag.

**Details:**

This macro read the bit indicating to the software that LINFlex has detected a falling edge on the LINRX pin in Sleep mode.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_UART_WUF_Read(offset)`

### 3.5.2.247 Macro LINFLEX\_LLM\_WAKEUP\_Clear

LINSR registers - Wake-up Flag.

**Details:**

This macro clears the Wake-up Flag.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_WAKEUP_Clear(offset)`

### 3.5.2.248 Macro LINFLEX\_LLM\_WAKEUP\_RECEIVED

LINSR registers - Wake-up Flag.

**Details:**

This macro read the bit indicating to the software that LINFlex has detected a falling edge on the LINRX pin.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_LLM_WAKEUP_RECEIVED(offset)`

### 3.5.2.249 Macro LINFLEX\_LLM\_WURQ\_Set

LINCR2 registers - Wake-up Generation Request.

**Details:**

Setting this bit generates a wake-up pulse.

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.



**Prototype:** `LINFLEX_LLM_WURQ_Set(offset)`

### 3.5.2.250 Macro LINFLEX\_UARTCR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register UART Mode Control Register (UARTCR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_UARTCR(offset)`

### 3.5.2.251 Macro LINFLEX\_UARTSR

LINFLEX Registers macro definition.

**Details:**

LINFLEX macro to access register UART Mode Status Register (UARTSR).

**Violates:** MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

**Prototype:** `LINFLEX_UARTSR(offset)`

## 3.5.3 Enum Reference

Enumeration of all constants supported by the Lin driver.

## 3.5.4 Function Reference

Functions supported by the Lin driver.

### 3.5.4.1 Function Lin\_CheckWakeup

Validates for upper layers the wake up of LIN channel.

**Details:**

This function identifies if the addressed LIN channel has been woken up by the LIN bus transceiver. This API is used when the LIN channel wake up functionality is disabled (wake up interrupt is disabled). It checks the wake up flag from the addressed LIN channel which must be in sleep mode and have the wake up functionality disabled.

**Implements:** DLIN07700, DLIN07701, DLIN07702, DLIN07703, DLIN07704, DLIN07705, DLIN07707, DLIN07708, DLIN07709, DLIN07710, DLIN07711

**Note**

Autosar Service ID: 0x0A.Synchronous, non reentrant function.

**Prototype:** Std\_ReturnType Lin\_CheckWakeup(uint8 Channel);

**Table 3-42. Lin\_CheckWakeup Return Values**

| Name     | Description                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------|
| E_NOT_OK | If the LIN Channel is not valid or LIN driver is not initialized or the addressed LIN Channel is not in sleep state. |
| E_OK     | Otherwise.                                                                                                           |

### 3.5.4.2 Function Lin\_GetStatus

Gets the status of the LIN driver.

**Details:**

This function returns the state of the current transmission, reception or operation status. If the reception of a Slave response was successful then this service provides a pointer to the buffer where the data is stored.

**Return:** Lin\_StatusType.

**Implements:** DLIN07750, DLIN07751, DLIN07752, DLIN07753, DLIN07754, DLIN07755, DLIN07756, DLIN07757, DLIN07758, DLIN07759, DLIN07760, DLIN07761, DLIN07762, DLIN07763

**Note**

Autosar Service ID: 0x08.Synchronous, non reentrant function.

**Prototype:** Lin\_StatusType Lin\_GetStatus(uint8 Channel, uint8 \*\*Lin\_SduPtr);

**Table 3-43. Lin\_GetStatus Arguments**

| Type     | Name       | Direction | Description                                                                                                                    |
|----------|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| uint8    | Channel    | input     | LIN channel to be checked.                                                                                                     |
| uint8 ** | Lin_SduPtr | output    | Lin_SduPtr pointer to pointer to a shadow buffer or memory mapped LIN Hardware receive buffer where the current SDU is stored. |

**Table 3-44. Lin\_GetStatus Return Values**

| Name                | Description                                                               |
|---------------------|---------------------------------------------------------------------------|
| LIN_NOT_OK          | Development or production error rised none of the below conditions.       |
| LIN_TX_OK           | Successful transmission.                                                  |
| LIN_TX_BUSY         | Ongoing transmission of header or response.                               |
| LIN_TX_HEADER_ERROR | Error occurred during header transmission.                                |
| LIN_TX_ERROR        | Error occurred during response transmission.                              |
| LIN_RX_OK           | Reception of correct response.                                            |
| LIN_RX_BUSY         | Ongoing reception where at least one byte has been received.              |
| LIN_RX_ERROR        | Error occurred during reception.                                          |
| LIN_RX_NO_REPONSE   | No data byte has been received yet.                                       |
| LIN_OPERATIONAL     | Channel is ready for next header. transmission and no data are available. |
| LIN_CH_SLEEP        | Channel is in sleep mode.                                                 |

### 3.5.4.3 Function Lin\_GetVersionInfo

Returns the version information of this module.

#### Details:

The version information includes:

- Two bytes for the Vendor ID
- Two bytes for the Module ID
- One byte for the Instance ID
- Three bytes version number. The numbering shall be vendor specific: it consists of:
  - The major, the minor and the patch version number of the module;
  - The AUTOSAR specification version number shall not be included. The AUTOSAR specification version number is checked during compile time and therefore not required in this API.

**Return:** void.

**Pre:** Preconditions as text description. Optional tag.

**Implements:** DLIN07100, DLIN07101, DLIN07102, DLIN07103, DLIN07104, DLIN07105, DLIN07106

### Note

Autosar Service ID: 0x01.Synchronous, non reentrant function.

**Prototype:** void Lin\_GetVersionInfo(Std\_VersionInfoType \*versioninfo);

**Table 3-45. Lin\_GetVersionInfo Arguments**

| Type                  | Name        | Direction     | Description                                                 |
|-----------------------|-------------|---------------|-------------------------------------------------------------|
| Std_VersionInfoType * | versioninfo | input, output | Pointer for storing the version information of this module. |

### 3.5.4.4 Function Lin\_GoToSleep

The service instructs the driver to transmit a go-to-sleep-command on the addressed LIN channel.

**Details:**

This function stops any ongoing transmission and initiates the transmission of the sleep command (master command frame with ID = 0x3C and data = (0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF)). State transition in LIN\_CH\_SLEEP\_STATE shall be done after the completion of the sleep command transmission regardless of the success (therefore the ISR is responsible to put the channel in LIN\_CH\_SLEEP\_STATE).

**Return:** Std\_ReturnType.

**Implements:** DLIN07800, DLIN07801, DLIN07802, DLIN07803, DLIN07804, DLIN07805, DLIN07806, DLIN07807, DLIN07808, DLIN07809, DLIN07810, DLIN07811

### Note

Autosar Service ID: 0x06.Synchronous, non reentrant function.

**Prototype:** Std\_ReturnType Lin\_GoToSleep(uint8 Channel);

**Table 3-46. Lin\_GoToSleep Arguments**

| Type  | Name    | Direction | Description                |
|-------|---------|-----------|----------------------------|
| uint8 | Channel | input     | LIN channel to be checked. |

**Table 3-47. Lin\_GoToSleep Return Values**

| Name     | Description                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------|
| E_NOT_OK | If the LIN Channel is not valid or LIN driver is not initialized or LIN Channel is in sleep state or a timeout occurs. |
| E_OK     | Otherwise.                                                                                                             |

### 3.5.4.5 Function Lin\_GoToSleepInternal

Put a Lin channel in the internal sleep state.

#### **Details:**

Stops any ongoing transmission, sets the channel state to LIN\_CH\_SLEEP and put the LIN hardware unit to a reduced power operation mode.

**Return:** Std\_ReturnType.

**Implements:** DLIN07300, DLIN07301, DLIN07302, DLIN07303, DLIN07304, DLIN07305, DLIN07306, DLIN07307, DLIN07308, DLIN07309, DLIN07310, DLIN07311, DLIN07312, DLIN07313, DLIN07314, DLIN07315, DLIN07316

#### **Note**

Autosar Service ID: 0x09.Synchronous, non reentrant function.

**Prototype:** Std\_ReturnType Lin\_GoToSleepInternal(uint8 Channel);

**Table 3-48. Lin\_GoToSleepInternal Arguments**

| Type  | Name    | Direction | Description                  |
|-------|---------|-----------|------------------------------|
| uint8 | Channel | input     | LIN channel to be addressed. |

**Table 3-49. Lin\_GoToSleepInternal Return Values**

| Name     | Description                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------|
| E_NOT_OK | If the LIN Channel is not valid or LIN driver is not initialized or LIN Channel is in sleep state or a timeout occurs. |
| E_OK     | Otherwise.                                                                                                             |

### 3.5.4.6 Function Lin\_Init

Initializes the LIN module.

#### Details:

This function performs software initialization of LIN driver:

- Clears the shadow buffer of all available Lin channels
- Set LIN channel state machine of all available Lin channels to LIN\_CH\_OPERATIONAL\_STATE
- Set frame operation state machine of all available LIN channels to LIN\_CH\_READY\_STATE
- Set driver state machine to LIN\_INIT.

**Return:** void.

**Implements:** DLIN07400, DLIN07401, DLIN07402, DLIN07403, DLIN07404, DLIN07405, DLIN07406, DLIN07407, DLIN07408, DLIN07409, DLIN07410, DLIN07411, DLIN09003

#### **Note**

Autosar Service ID: 0x00.Synchronous, non reentrant function.

**Prototype:** void Lin\_Init(const Lin\_ConfigType \*Config);

**Table 3-50. Lin\_Init Arguments**

| Type                 | Name   | Direction | Description                              |
|----------------------|--------|-----------|------------------------------------------|
| constLin_ConfigType* | Config | input     | Pointer to LIN driver configuration set. |

### 3.5.4.7 Function Lin\_SendFrame

Sends a LIN frame.

#### Details:

Sends a LIN header and a LIN response, if necessary. The direction of the frame response (master response, slave response, slave-to-slave communication) is provided by the PduInfoPtr.

**Return:** Std\_ReturnType.

**Implements:** DLIN07500, DLIN07501, DLIN07502, DLIN07503, DLIN07504, DLIN07505, DLIN07506, DLIN07507, DLIN07508, DLIN07509, DLIN07510, DLIN07511, DLIN07512, DLIN07513, DLIN07514, DLIN07515, DLIN07516, DLIN07517, DLIN07518, DLIN07519, DLIN07520, DLIN07521, DLIN07522

### Note

Autosar Service ID: 0x04.Synchronous, non reentrant function.

**Prototype:** Std\_ReturnType Lin\_SendFrame(uint8 Channel, Lin\_PduType \*PduInfoPtr);

**Table 3-51. Lin\_SendFrame Arguments**

| Type         | Name       | Direction | Description                                                                                |
|--------------|------------|-----------|--------------------------------------------------------------------------------------------|
| uint8        | Channel    | input     | LIN channel to be addressed.                                                               |
| Lin_PduType* | PduInfoPtr | input     | Pointer to PDU containing the PID, Checksum model, Response type, DI and SDU data pointer. |

**Table 3-52. Lin\_SendFrame Return Values**

| Name     | Description                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| E_NOT_OK | If the LIN Channel is not valid or LIN driver is not initialized or PduInfoPtr is NULL or a timeout occurs or LIN Channel is in sleep state. |
| E_OK     | Otherwise.                                                                                                                                   |

### 3.5.4.8 Function Lin\_Wakeup

Generates a wake up pulse.

#### **Details:**

This function shall sent a wake up signal to the LIN bus and put the LIN channel in LIN\_CH\_OPERATIONAL\_STATE state.

**Return:** Std\_ReturnType.

**Implements:** DLIN07600, DLIN07601, DLIN07602, DLIN07603, DLIN07604, DLIN07605, DLIN07606, DLIN07607, DLIN07608, DLIN07609, DLIN07610, DLIN07611

### Note

Autosar Service ID: 0x07.Synchronous, non reentrant function.

**Prototype:** Std\_ReturnType Lin\_Wakeup(uint8 Channel);

Table 3-53. Lin\_Wakeup Arguments

| Type  | Name    | Direction | Description                  |
|-------|---------|-----------|------------------------------|
| uint8 | Channel | input     | LIN channel to be addressed. |

Table 3-54. Lin\_Wakeup Return Values

| Name     | Description                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------|
| E_NOT_OK | If the LIN driver is not in sleep state or LIN Channel is not valid or LIN driver is not initialized. |
| E_OK     | Otherwise.                                                                                            |

3.5.5 Structs Reference

Data structures supported by the Lin driver.

3.5.5.1 Structure Lin\_ChannelConfigType

LIN channel configuration type structure.

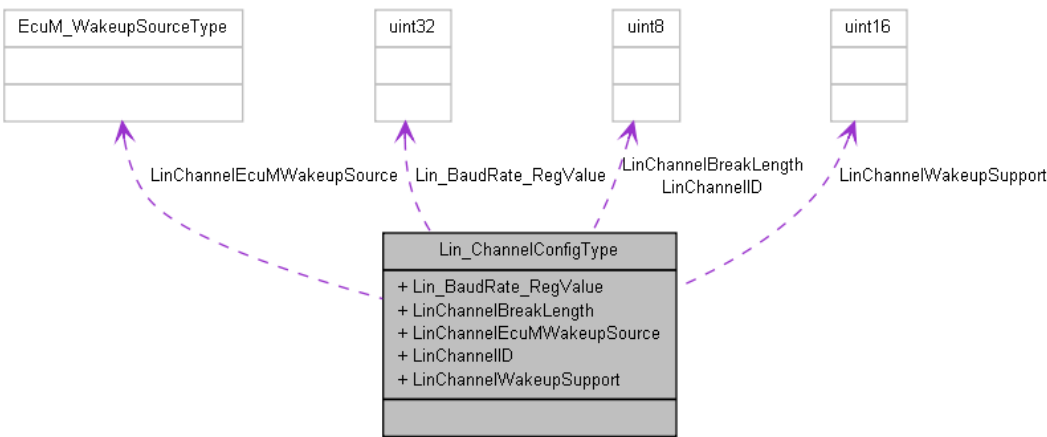


Figure 3-1. Struct Lin\_ChannelConfigType

**Details:**

This is the type of the external data structure containing the overall initialization data for one LIN Channel. A pointer to such a structure is provided to the LIN channel initialization routine for configuration of the LIN hardware channel.



**Implements:** DLIN03013**Declaration:**

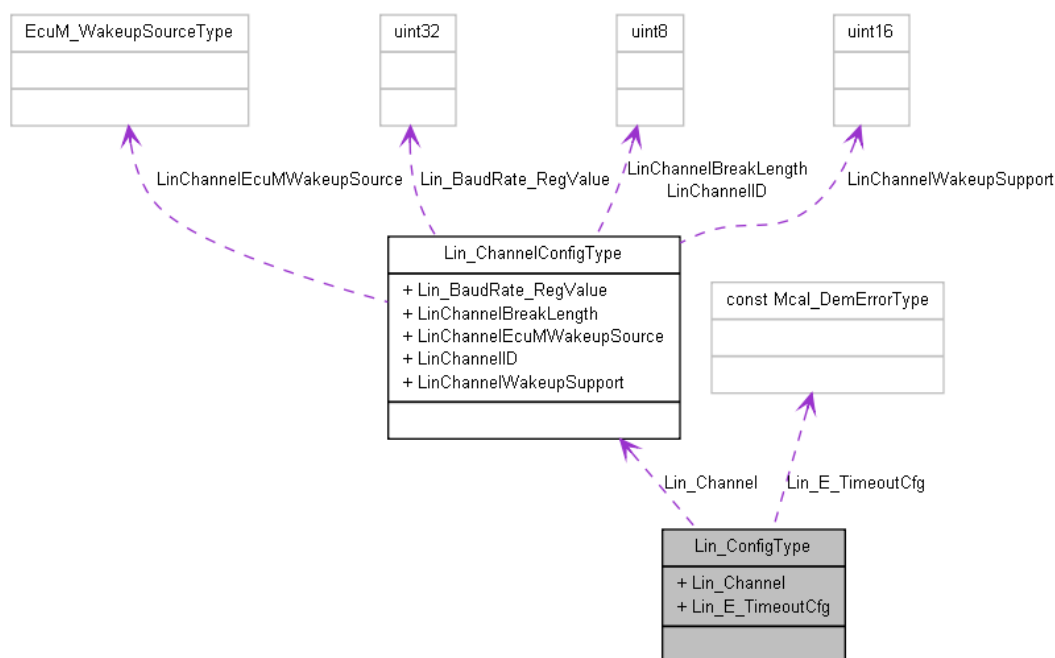
```
typedef struct
{
    uint32 Lin_BaudRate_RegValue,
    uint8 LinChannelBreakLength,
    EcuM_WakeupSourceType LinChannelEcuMWakeupSource,
    uint8 LinChannelID,
    uint16 LinChannelWakeupSupport
} Lin_ChannelConfigType;
```

**Table 3-55. Structure Lin\_ChannelConfigType member description**

| Member                     | Description |
|----------------------------|-------------|
| Lin_BaudRate_RegValue      |             |
| LinChannelBreakLength      |             |
| LinChannelEcuMWakeupSource |             |
| LinChannelID               |             |
| LinChannelWakeupSupport    |             |

**3.5.5.2 Structure Lin\_ConfigType**

LIN driver configuration type structure.

**Figure 3-2. Struct Lin\_ConfigType****Details:**

This is the type of the pointer to the external data LIN Channels. A pointer of such a structure is provided to the LIN driver initialization routine for configuration of the LIN hardware channel.

**Implements:** DLIN03014

**Declaration:**

```
typedef struct
{
    constLin_ChannelConfigType*const
Lin_Channel[LIN_HW_MAX_MODULES],
    const Mcal_DemErrorType Lin_E_TimeoutCfg
} Lin_ConfigType;
```

**Table 3-56. Structure Lin\_ConfigType member description**

| Member           | Description       |
|------------------|-------------------|
| Lin_Channel      | Hardware channel. |
| Lin_E_TimeoutCfg |                   |

## 3.5.6 Types Reference

Types structures supported by the Lin driver.

### 3.5.6.1 Typedef Lin\_NotifyType

LIN channel physical error notification callback.

**Implements:**

**Type:** void(

## 3.5.7 Variables Reference

Variables supported by the Lin driver.

### 3.5.7.1 Variable Lin\_LinChannel\_0\_C0

Configuration for Lin Channel 0 - Configuration: LinGlobalConfig\_0.

**Implements:** DLIN03303

**Violates:** include statements in a file should only be preceded by other preprocessor directives or comments.

**Implements:** DLIN03603

**Violates:** All declarations and definitions of objects or functions at file scope shall have internal linkage unless external linkage is required.

**Declaration:**

```
const Lin_ChannelConfigType Lin_LinChannel_0_C0
```

### 3.5.7.2 Variable Lin\_LinChannel\_1\_C0

Configuration for Lin Channel 1 - Configuration: LinGlobalConfig\_0.

**Implements:** DLIN03303

**Implements:** DLIN03603

**Violates:** All declarations and definitions of objects or functions at file scope shall have internal linkage unless external linkage is required.

**Declaration:**

```
const Lin_ChannelConfigType Lin_LinChannel_1_C0
```

### 3.5.7.3 Variable LinGlobalConfig\_0

Lin Configuration data for LinGlobalConfig\_0.

**Violates:** Precautions shall be taken in order to prevent the contents of a header file being included twice.

**Implements:** DLIN03304

**Implements:** DLIN03602

**Violates:** All declarations and definitions of objects or functions at file scope shall have internal linkage unless external linkage is required.

## **Declaration:**

```
const Lin_ConfigType LinGlobalConfig_0
```

## Chapter 4

# Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the Lin Driver. The most of the parameters are described below.

### 4.1 Configuration elements of Lin

Included forms :

- IMPLEMENTATION\_CONFIG\_VARIANT
- NonAutosar
- LinGeneral
- CommonPublishedInformation
- LinGlobalConfig

Table 4-1. Revision table

| Revision | Date       |
|----------|------------|
| 4.1.0    | 2013-04-08 |

### 4.2 Form IMPLEMENTATION\_CONFIG\_VARIANT

VariantPreCompile: Only precompile time configuration parameters.

VariantPostBuild: Mix of precompile and postbuild time configuration parameters.

If Config Variant = VariantPreCompile, the files Lin\_Cfg.h and Lin\_Cfg.c shall be used.

If Config Variant = VariantPostBuild, the files Lin\_Cfg.h and Lin\_PBcfg.c shall be used.



Figure 4-1. Tresos Plugin snapshot for IMPLEMENTATION\_CONFIG\_VARIANT form.

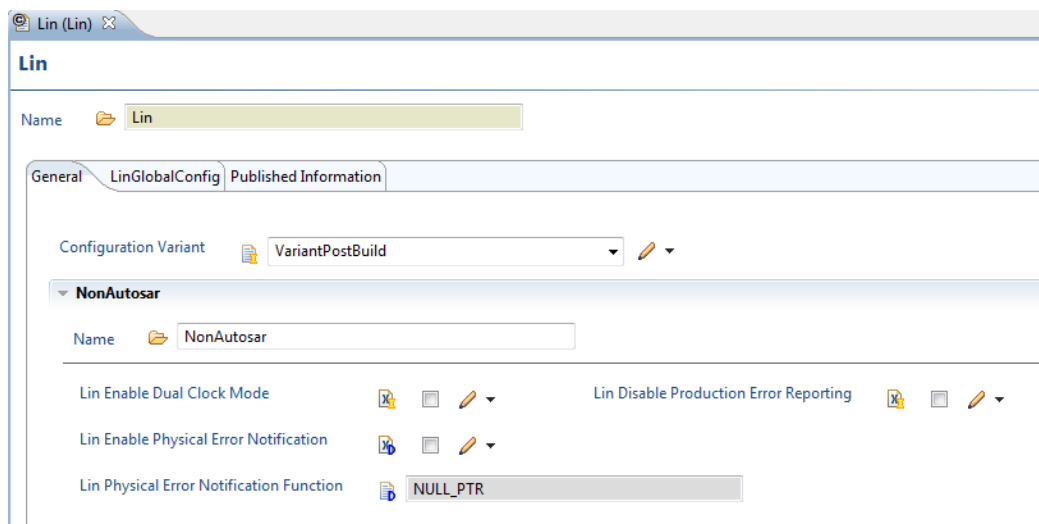
**Table 4-2. Attribute IMPLEMENTATION\_CONFIG\_VARIANT detailed description**

| Property      | Value                                 |
|---------------|---------------------------------------|
| Label         | Configuration Variant                 |
| Type          | ENUMERATION                           |
| Symbolic Name | false                                 |
| Default       | VariantPostBuild                      |
| Range         | VariantPostBuild<br>VariantPreCompile |

## 4.3 Form NonAutosar

### NonAutosar

This container contains the global configuration parameters of the Non-Autosar Lin driver. This container is a MultipleConfigurationContainer, i.e. this container and its sub-containers exist once per configuration set.

**Figure 4-2. TRESOS Plugin snapshot for NonAutosar form.**

### 4.3.1 LinEnableDualClockMode (NonAutosar)

#### LinEnableDualClockMode

Switches the Dual Clock Mode ON or OFF.

**Table 4-3. Attribute LinEnableDualClockMode (NonAutosar) detailed description**

| Property      | Value                      |
|---------------|----------------------------|
| Label         | Lin Enable Dual Clock Mode |
| Type          | BOOLEAN                    |
| Origin        | Custom                     |
| Symbolic Name | false                      |
| Default       | false                      |

### 4.3.2 LinDisableDemReportErrorStatus (NonAutosar)

#### LinDisableDemReportErrorStatus

Switches the Diagnostic Error Reporting and Notification OFF.

**Table 4-4. Attribute LinDisableDemReportErrorStatus (NonAutosar) detailed description**

| Property      | Value                                  |
|---------------|----------------------------------------|
| Label         | Lin Disable Production Error Reporting |
| Type          | BOOLEAN                                |
| Origin        | Custom                                 |
| Symbolic Name | false                                  |
| Default       | false                                  |

### 4.3.3 LinEnablePhyiscalErrorNotification (NonAutosar)

#### LinEnablePhyiscalErrorNotification

Switches the Physical Error Notification ON or OFF.

**Table 4-5. Attribute LinEnablePhyiscalErrorNotification (NonAutosar) detailed description**

| Property      | Value                                  |
|---------------|----------------------------------------|
| Label         | Lin Enable Physical Error Notification |
| Type          | BOOLEAN                                |
| Origin        | Custom                                 |
| Symbolic Name | false                                  |
| Default       | false                                  |

### 4.3.4 LinPhysicalErrorNotificationFunction (NonAutosar)

This parameter is a reference to a notification function.

**Table 4-6. Attribute LinPhysicalErrorNotificationFunction (NonAutosar) detailed description**

| Property      | Value                                    |
|---------------|------------------------------------------|
| Label         | Lin Physical Error Notification Function |
| Type          | FUNCTION-NAME                            |
| Origin        | Custom                                   |
| Symbolic Name | false                                    |
| Default       | NULL_PTR                                 |

## 4.4 Form LinGeneral

### LinGeneral

Autosar Requirements: LIN183\_Conf

This container contains the parameters related to each LIN Driver Unit. This container is a MultipleConfigurationContainer, i.e. this container and its sub-containers exit once per configuration set.



Lin (Lin) X

Lin

Name Lin

General LinGlobalConfig Published Information

Config Variant VariantPostBuild

▼ NonAutosar

NonAutosar

Lin Enable Dual Clock Mode ☐

▼ LinGeneral

LinGeneral

Lin Development Error Detect ☒

InstanceId (0 -> 255) 0

Lin Timeout Duration (1 -> 2147483648) 1000

Lin VersionInfo Api ☒

**Figure 4-3. Tresos Plugin snapshot for LinGeneral form.**

## 4.4.1 LinDevErrorDetect (LinGeneral)

### LinDevErrorDetect

Autosar Requirements: LIN066\_Conf

Switches the Development Error Detection and Notification ON or OFF.

**Table 4-7. Attribute LinDevErrorDetect (LinGeneral) detailed description**

| Property      | Value                           |
|---------------|---------------------------------|
| Label         | Lin Development Error Detection |
| Type          | BOOLEAN                         |
| Origin        | AUTOSAR_ECUC                    |
| Symbolic Name | false                           |
| Default       | true                            |

## 4.4.2 LinIndex (LinGeneral)

### LinIndex

Autosar Requirements: LIN179\_Conf

Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

Note, this parameter is not used in the current implementation.

**Table 4-8. Attribute LinIndex (LinGeneral) detailed description**

| Property      | Value                                                       |
|---------------|-------------------------------------------------------------|
| Label         | InstanceId                                                  |
| Type          | INTEGER                                                     |
| Origin        | AUTOSAR_ECUC                                                |
| Symbolic Name | false                                                       |
| Default       | 0                                                           |
| Invalid       | Range<br><div> <div>&lt;=255</div> <div>&gt;=0</div> </div> |

## 4.4.3 LinTimeoutDuration (LinGeneral)

### LinTimeoutDuration

Autosar Requirements: LIN093\_Conf

Specifies the maximum number of loops for blocking function until a timeout is raised in short term wait loops

**Table 4-9. Attribute LinTimeoutDuration (LinGeneral) detailed description**

| Property      | Value                                                         |
|---------------|---------------------------------------------------------------|
| Label         | Lin Timeout Duration                                          |
| Type          | INTEGER                                                       |
| Origin        | AUTOSAR_ECUC                                                  |
| Symbolic Name | false                                                         |
| Default       | 1000                                                          |
| Invalid       | Range<br><div> <div>&lt;=65535</div> <div>&gt;=0</div> </div> |

#### 4.4.4 LinVersionInfoApi (LinGeneral)

##### LinVersionInfoApi

Autosar Requirements: LIN067\_Conf

Switches the Lin\_GetVersionInfo function ON or OFF.

**Table 4-10. Attribute LinVersionInfoApi (LinGeneral) detailed description**

| Property      | Value                       |
|---------------|-----------------------------|
| Label         | Provide Lin VersionInfo Api |
| Type          | BOOLEAN                     |
| Origin        | AUTOSAR_ECUC                |
| Symbolic Name | false                       |
| Default       | true                        |

## 4.5 Form CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

General
LinGlobalConfig
Published Information

CommonPublishedInformation

AUTOSAR Major Version
1

AUTOSAR Minor Version
2

AUTOSAR Patch Version
1

Numeric Module ID
82

Software Major Version
1

Software Minor Version
0

Software Patch Version
0

Vendor Api Infix

Vendor ID
43

Figure 4-4. Tresos Plugin snapshot for CommonPublishedInformation form.

### 4.5.1 ArReleaseMajorVersion (CommonPublishedInformation)

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

**Table 4-11. Attribute ArReleaseMajorVersion (CommonPublishedInformation) detailed description**

| Property      | Value                                                     |
|---------------|-----------------------------------------------------------|
| Label         | AUTOSAR Major Version                                     |
| Type          | INTEGER_LABEL                                             |
| Origin        | Custom                                                    |
| Symbolic Name | false                                                     |
| Default       | 4                                                         |
| Invalid       | Range<br><div> <div>&gt;=4</div> <div>&lt;=4</div> </div> |

### 4.5.2 ArReleaseMinorVersion (CommonPublishedInformation)

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

**Table 4-12. Attribute ArReleaseMinorVersion (CommonPublishedInformation) detailed description**

| Property      | Value                                                     |
|---------------|-----------------------------------------------------------|
| Label         | AUTOSAR Minor Version                                     |
| Type          | INTEGER_LABEL                                             |
| Origin        | Custom                                                    |
| Symbolic Name | false                                                     |
| Default       | 0                                                         |
| Invalid       | Range<br><div> <div>&gt;=0</div> <div>&lt;=0</div> </div> |

### 4.5.3 ArReleaseRevisionVersion (CommonPublishedInformation)

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

**Table 4-13. Attribute ArReleaseRevisionVersion (CommonPublishedInformation) detailed description**

| Property      | Value                                                     |
|---------------|-----------------------------------------------------------|
| Label         | AUTOSAR Release Revision Version                          |
| Type          | INTEGER_LABEL                                             |
| Origin        | Custom                                                    |
| Symbolic Name | false                                                     |
| Default       | 3                                                         |
| Invalid       | Range<br><div> <div>&gt;=3</div> <div>&lt;=3</div> </div> |

### 4.5.4 ModuleId (CommonPublishedInformation)

Module ID of this module from Module List.

**Table 4-14. Attribute ModuleId (CommonPublishedInformation) detailed description**

| Property      | Value                 |
|---------------|-----------------------|
| Label         | Numeric Module ID     |
| Type          | INTEGER_LABEL         |
| Origin        | Custom                |
| Symbolic Name | false                 |
| Default       | 82                    |
| Invalid       | Range<br>>=82<br><=82 |

### 4.5.5 SwMajorVersion (CommonPublishedInformation)

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

**Table 4-15. Attribute SwMajorVersion (CommonPublishedInformation) detailed description**

| Property      | Value                  |
|---------------|------------------------|
| Label         | Software Major Version |
| Type          | INTEGER_LABEL          |
| Origin        | Custom                 |
| Symbolic Name | false                  |
| Default       | 1                      |
| Invalid       | Range<br>>=1<br><=1    |

### 4.5.6 SwMinorVersion (CommonPublishedInformation)

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

**Table 4-16. Attribute SwMinorVersion (CommonPublishedInformation) detailed description**

| Property | Value                  |
|----------|------------------------|
| Label    | Software Minor Version |
| Type     | INTEGER_LABEL          |

*Table continues on the next page...*

**Table 4-16. Attribute SwMinorVersion (CommonPublishedInformation) detailed description (continued)**

| Property      | Value                                        |
|---------------|----------------------------------------------|
| Origin        | Custom                                       |
| Symbolic Name | false                                        |
| Default       | 0                                            |
| Invalid       | Range<br><div>&gt;=0</div> <div>&lt;=0</div> |

### 4.5.7 SwPatchVersion (CommonPublishedInformation)

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

**Table 4-17. Attribute SwPatchVersion (CommonPublishedInformation) detailed description**

| Property      | Value                                        |
|---------------|----------------------------------------------|
| Label         | Software Patch Version                       |
| Type          | INTEGER_LABEL                                |
| Origin        | Custom                                       |
| Symbolic Name | false                                        |
| Default       | 1                                            |
| Invalid       | Range<br><div>&gt;=1</div> <div>&lt;=1</div> |

### 4.5.8 VendorApiInfix (CommonPublishedInformation)

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name. This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

<ModuleName>\_>VendorId>\_<VendorApiInfix><Api name from SWS>. E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can\_Write defined in the SWS will translate to Can\_123\_v11r456Write. This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

**Table 4-18. Attribute VendorApiInfix (CommonPublishedInformation) detailed description**

| Property      | Value            |
|---------------|------------------|
| Label         | Vendor Api Infix |
| Type          | STRING_LABEL     |
| Origin        | Custom           |
| Symbolic Name | false            |
| Default       |                  |
| Enable        | false            |

### 4.5.9 VendorId (CommonPublishedInformation)

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

**Table 4-19. Attribute VendorId (CommonPublishedInformation) detailed description**

| Property      | Value                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------|
| Label         | Vendor ID                                                                                       |
| Type          | INTEGER_LABEL                                                                                   |
| Origin        | Custom                                                                                          |
| Symbolic Name | false                                                                                           |
| Default       | 43                                                                                              |
| Invalid       | Range<br><div style="margin-left: 20px;"> <math>\geq 43</math><br/> <math>\leq 43</math> </div> |

## 4.6 Form LinGlobalConfig

This container contains the global configuration parameter of the Lin driver. This container is a MultipleConfigurationContainer, i.e. this container and its sub-containers exist once per configuration set.



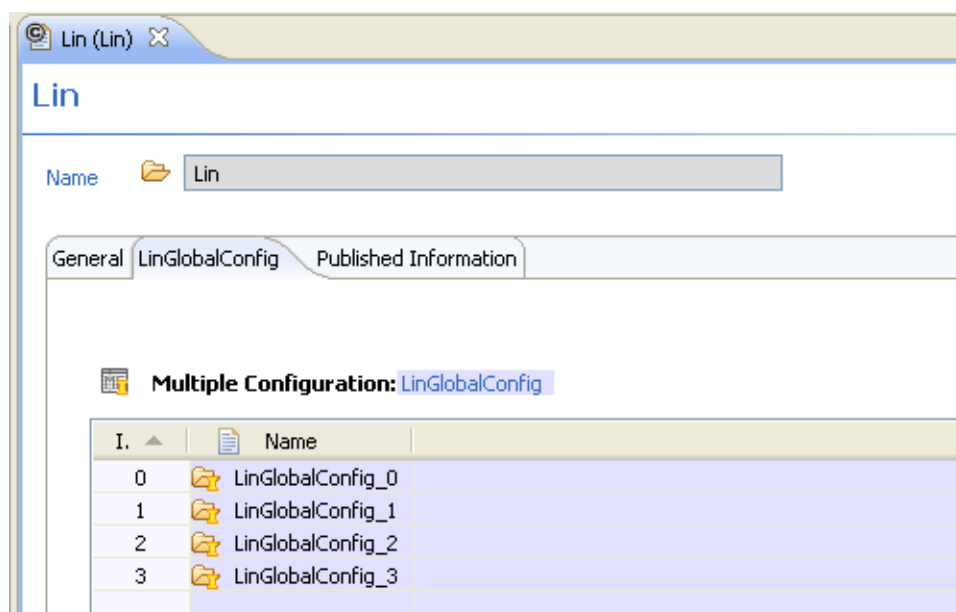


Figure 4-5. TresoS Plugin snapshot for LinGlobalConfig form.

### 4.6.1 Form LinChannel

#### Note

This container contains the configuration (parameters) of the LIN Controller(s).

"User should use unique names for naming the LIN channels across different LinGlobalConfig Sets."

**Is included by form :** [Form LinGlobalConfig](#)

LinChannel

Name LinChannel\_0

General

Lin Channel ID (0 -> 9) 0

Lin hardware channel LinHWCh\_0

LinClockRef /Mcu/Mcu/McuModuleConfiguratio\_0/McuClockSettingCon

LinClockRef\_Alternate /Mcu/Mcu/McuModuleConfiguratio\_0/McuClockSettingCon

Lin Channel BaudRate (bps) (1000 -> 20000) 2400

Break Length (bits) BL\_15

Lin Channel Wake UP support

EcuM WakeUP source /EcuM/EcuM/EcuMConfiguration\_0/EcuMWakeupSource\_c

Figure 4-6. Tresos Plugin snapshot for LinChannel form.

4.6.1.1 LinChannelId (LinChannel)

Identifies the LIN channel. Replaces LIN\_CHANNEL\_INDEX\_NAME from the LIN SWS.

Table 4-20. Attribute LinChannelId (LinChannel) detailed description

| Property      | Value                |
|---------------|----------------------|
| Label         | Lin Channel ID       |
| Type          | INTEGER              |
| Origin        | AUTOSAR_ECUC         |
| Symbolic Name | true                 |
| Invalid       | Range<br>>=0<br><=10 |

4.6.1.2 LinChannelBaudRate (LinChannel)

LinChannelBaudRate

Autosar Requirements: LIN180\_Conf

Specifies the baud rate of the LIN channel in 'bps'. Valid range: 1000..20000.

**Table 4-21. Attribute LinChannelBaudRate (LinChannel) detailed description**

| Property      | Value                      |
|---------------|----------------------------|
| Label         | Lin Channel BaudRate (bps) |
| Type          | INTEGER                    |
| Origin        | AUTOSAR_ECUC               |
| Symbolic Name | false                      |
| Default       | 9600                       |
| Invalid       | Range<br><=20000<br>>=1000 |

#### 4.6.1.3 BreakLength (LinChannel)

##### Note

Defines the break length in bits.

This Parameter is an Implementation Specific Parameter.

**Table 4-22. Attribute BreakLength (LinChannel) detailed description**

| Property      | Value                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Label         | Break Length (bits)                                                                                                                          |
| Type          | ENUMERATION                                                                                                                                  |
| Origin        | Custom                                                                                                                                       |
| Symbolic Name | false                                                                                                                                        |
| Default       | BL_13                                                                                                                                        |
| Range         | BL_10<br>BL_11<br>BL_12<br>BL_13<br>BL_14<br>BL_15<br>BL_16<br>BL_17<br>BL_18<br>BL_19<br>BL_20<br>BL_21<br>BL_22<br>BL_23<br>BL_36<br>BL_50 |

#### 4.6.1.4 LinHwChannel (LinChannel)

##### Note

Selects the physical LIN Channel.

This Parameter is an Implementation Specific Parameter.

**Table 4-23. Attribute LinHwChannel (LinChannel) detailed description**

| Property      | Value                |
|---------------|----------------------|
| Label         | Lin hardware channel |
| Type          | ENUMERATION          |
| Origin        | Custom               |
| Symbolic Name | false                |

#### 4.6.1.5 LinChannelWakeupSupport (LinChannel)

##### LinChannelWakeupSupport

Autosar Requirements: LIN182\_Conf

Specifies if the LIN hardware channel supports wake up functionality.

**Table 4-24. Attribute LinChannelWakeupSupport (LinChannel) detailed description**

| Property      | Value                       |
|---------------|-----------------------------|
| Label         | Lin Channel Wake UP support |
| Type          | BOOLEAN                     |
| Origin        | AUTOSAR_ECUC                |
| Symbolic Name | false                       |
| Default       | false                       |

#### 4.6.1.6 LinClockRef (LinChannel)

Reference to the LIN clock source configuration, which is set in the MCU driver configuration.

**Table 4-25. Attribute LinClockRef (LinChannel) detailed description**

| Property | Value        |
|----------|--------------|
| Type     | REFERENCE    |
| Origin   | AUTOSAR_ECUC |

#### 4.6.1.7 LinClockRef\_Alternate (LinChannel)

Alternate reference to the LIN clock source configuration, which is set in the MCU driver configuration, used in Low Power Mode.

**Table 4-26. Attribute LinClockRef\_Alternate (LinChannel) detailed description**

| Property | Value     |
|----------|-----------|
| Type     | REFERENCE |
| Origin   | Custom    |

#### 4.6.1.8 LinChannelEcuMWakeupSource (LinChannel)

**Table 4-27. Attribute LinChannelEcuMWakeupSource (LinChannel) detailed description**

| Property | Value                   |
|----------|-------------------------|
| Label    | EcuM WakeUP source      |
| Type     | SYMBOLIC-NAME-REFERENCE |
| Origin   | AUTOSAR_ECUC            |
| Enable   | true                    |



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